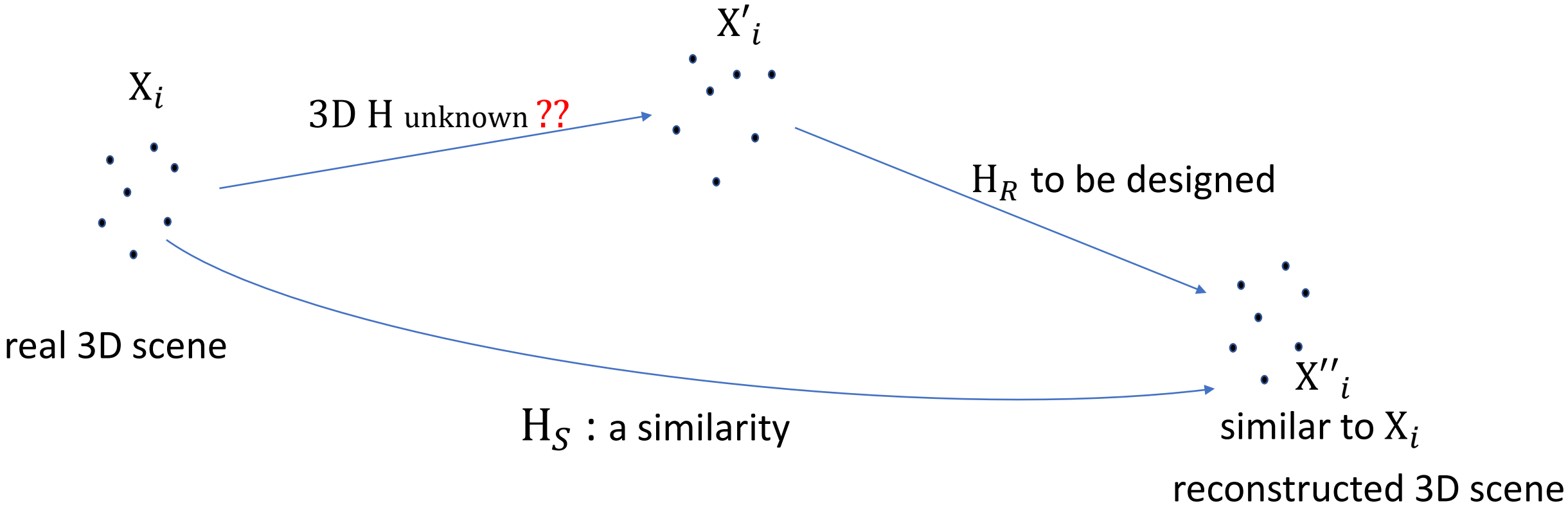
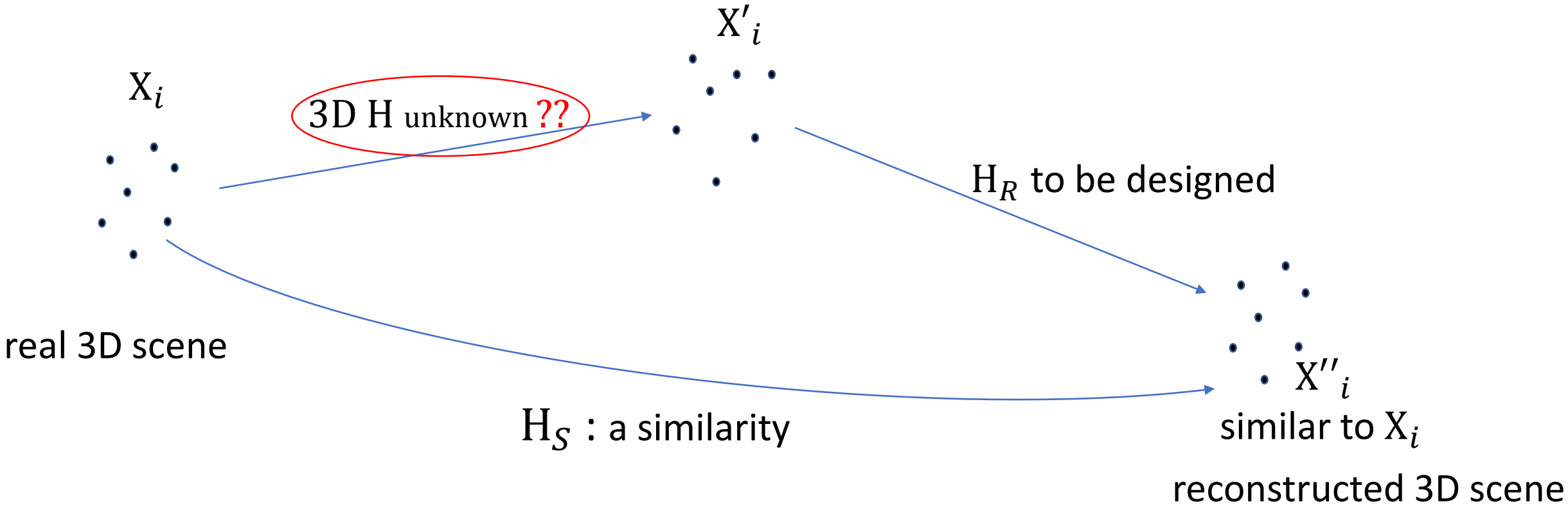
3D Reconstruction

3D shape reconstruction from images



3D shape reconstruction from images



3D reconstruction problem

A step was left behind, while studying 3D projective Geometry:

- How to construct an observation of the 3D scene, that is a 3D projectivity of it?
- Take 2D images of 3D scene points X_i and find correspondences $\mathrm{x}_1^i \leftrightarrow \mathrm{x}_2^i$
- Find fundamental matrix(es) F_{12} by solving $\begin{cases} \mathrm{x}_2^{Ti}\mathrm{F}_{12}\mathrm{x}_1^i = 0 \\ i = 1 \dots N \end{cases}$ for F_{12}
- Take a tentative pair of cameras compatible with F_{12}

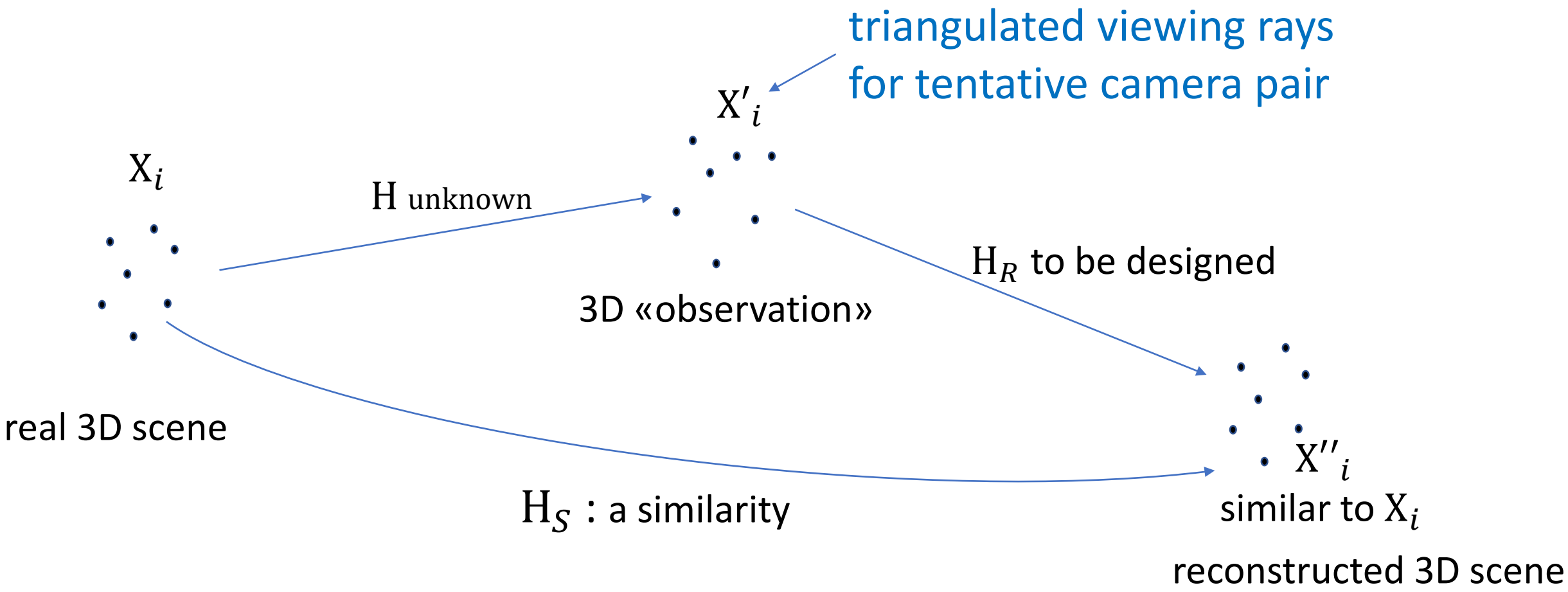
$$\mathrm{P}_1 = [\mathrm{I} \mid 0] \quad \mathrm{P}_2 = [[\mathrm{e}_2]_\times \mathrm{F}_{12} + \mathrm{e}_2 \mathrm{v}^{\mathrm{T}} \mid \lambda \mathrm{e}_2]$$

where e_2 is the LNS of F_{12}, λ and v are any nonzero scalar and vector

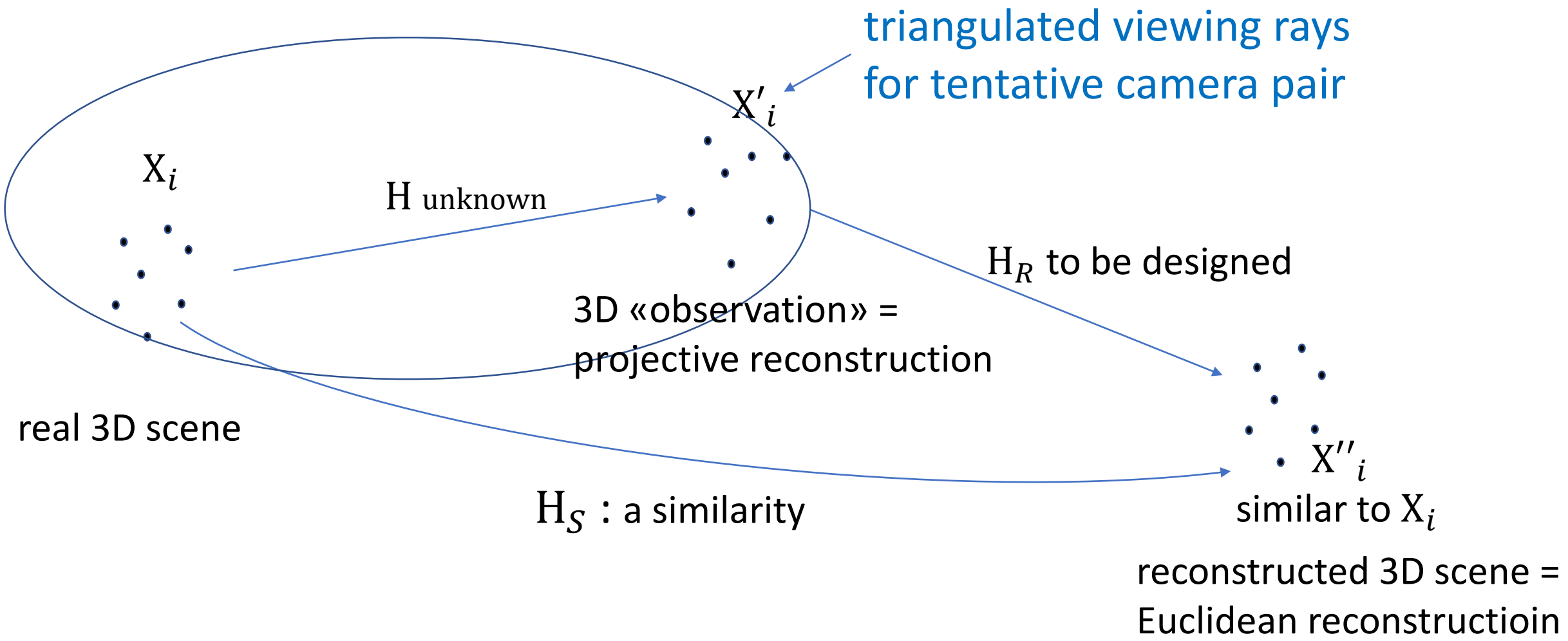
- Triangulate the pairs of viewing rays associated to $\mathrm{x}_1^i \leftrightarrow \mathrm{x}_2^i$ through cameras $\mathrm{P}_1\,\mathrm{P}_2$
- Take $X_i{}'$ as the points resulting from the triangulations:

THESE $X_i{}'$ ARE AN UNKNOWN PROJECTIVE TRANSFORMATION OF THE TRUE 3D POINTS X_i

3D shape reconstruction from images

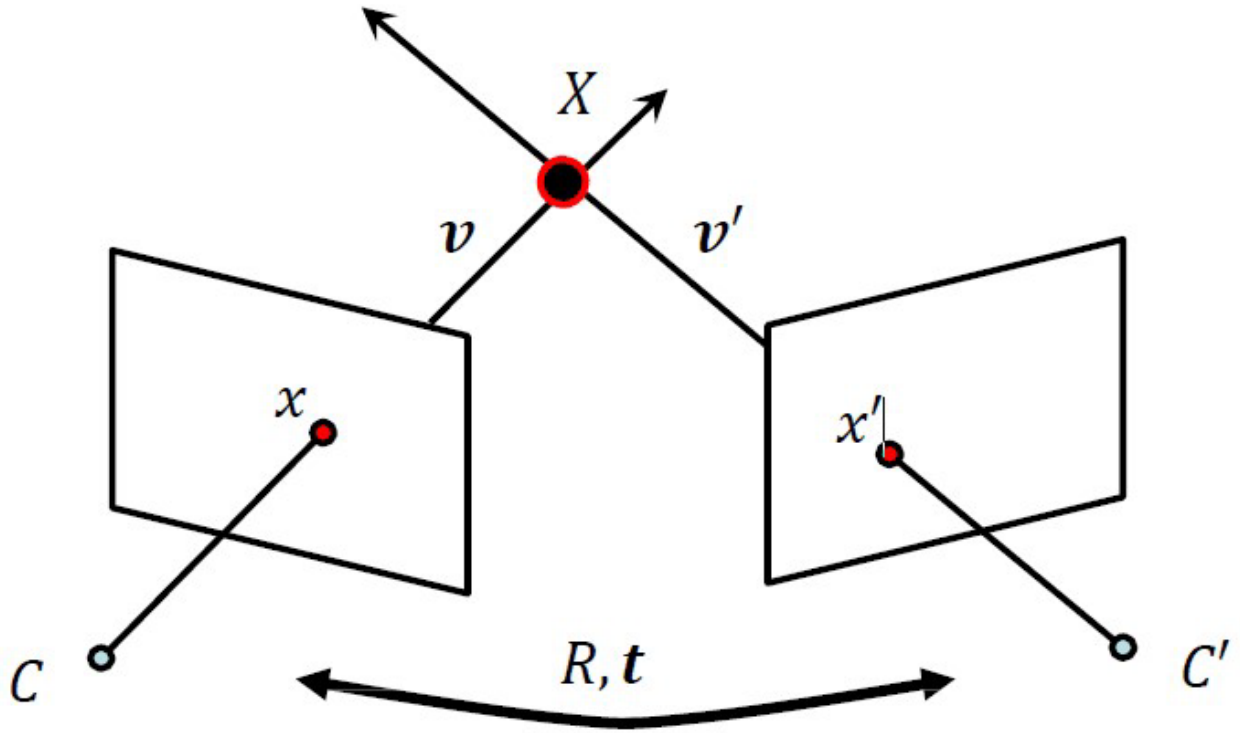


3D shape reconstruction: first step



scenario 2: **calibrated** structure from motion

x ←→x’ known; **R,t unknown**; K and K’ known;



use **epipolar constraint** to estimate R, t; → compute viewing rays v, v’

→ triangulation: $\boldsymbol{X} = \boldsymbol{v} \cap \boldsymbol{v}'$

Yet another expression of P

$$P = [\mathrm{M} \quad \mathrm{m}] \quad O = \begin{bmatrix} o \\ 1 \end{bmatrix} = \mathrm{RNS(P)}$$

$$\mathrm{M} = \mathrm{KR}_{cam \to world} \quad o = t_{world \to cam}$$

From

$$PO = [\mathrm{M} \quad \mathrm{m}] \begin{bmatrix} t_{world \to cam} \\ 1 \end{bmatrix} = \mathrm{M}t_{world \to cam} + \mathrm{m} = 0$$

is $m = -\mathrm{M}t_{world \to cam} = -\mathrm{KR}_{cam \to world} t_{world \to cam}$

But since

$$t_{cam \to world} = -\mathrm{R}_{cam \to world} t_{world \to cam},$$

Then $m = K t_{cam \to world}$

Hence $P = [\mathrm{M} \quad \mathrm{m}] = [K\mathrm{R}_{cam \to world} \quad K t_{cam \to world}]$

$$P = K[R \quad t]$$

where $R \stackrel{\text{def}}{=} \mathrm{R}_{cam \to world}$ and $t \stackrel{\text{def}}{=} t_{cam \to world}$

3D shape reconstruction from calibrated images

Calibrated cameras observe a same scene from unknown relative positions

Change image coordinates

$$\mathrm{x}_1 = \mathrm{K}_1\hat{\mathrm{x}}_1$$
$$\mathrm{x}_2 = \mathrm{K}_2\hat{\mathrm{x}}_2$$

$$\mathrm{x}_1 = P_1X \rightarrow K_1\hat{\mathrm{x}}_1 = K_1[\mathrm{R}_1 \quad \mathrm{t}_1]X \quad \rightarrow \quad \hat{\mathrm{x}}_1 = [\mathrm{R}_1 \quad \mathrm{t}_1]X$$

and, similarly, $\hat{\mathrm{x}}_2 = [\mathrm{R}_2 \quad \mathrm{t}_2]X$

Place world reference = camera-1 reference: $R_1 = I$ and $t_1 = 0$ (and call $R \stackrel{\text{def}}{=} \mathrm{R}_2$, $t \stackrel{\text{def}}{=} t_2$)

$$\hat{\mathrm{x}}_1 = \underbrace{[I \quad 0]}_{\hat{P}_1}X \text{ and } \hat{\mathrm{x}}_2 = \underbrace{[R \quad t]}_{\hat{P}_2}X$$

Essential matrix of two calibrated images

(Longuet-Higgins 1980)

$$\hat{\mathrm{x}}_2{}^T E_{12} \hat{\mathrm{x}}_1 = 0$$

$$E_{12} = [\hat{\mathrm{e}}_2]_\times \hat{\mathrm{M}}_2 \hat{\mathrm{M}}_1^{-1}$$

The epipole is the image projection of the center of the first camera center (the origin) onto the second camera $\hat{P}_2 = [R \quad t] : \hat{\mathrm{e}}_2 = [R \quad t] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = t$

thus $\boxed{E_{12} = [t]_\times \mathrm{R}}$

Relationship between Fundamental matrix F and Essential matrix E

$$\mathrm{x}_2^T F_{12} \mathrm{x}_1 = \hat{\mathrm{x}}_2{}^T \mathrm{K}_2^T F_{12} \mathrm{K}_1 \hat{\mathrm{x}}_1 = 0 = \hat{\mathrm{x}}_2{}^T E_{12} \hat{\mathrm{x}}_1 \longrightarrow \mathrm{K}_2^T F_{12} \mathrm{K}_1 = E_{12}$$

First derive R, and then triangulate

- **Property**: any 3x3 skew-symmetric matrix S can be written as

$$S = hU\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}U^T = hUZU^T \quad \text{where } \mathrm{Z} \triangleq \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } U \text{ is an orthogonal matrix}$$

Let $W \triangleq \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ be another orthogonal matrix: → it is $Z = \pm\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}W$

Thus, since matrix $[t]_\times$ is skew-symmetric, $[t]_\times = \pm UZU^T = \pm U\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}WU^T$

Therefore $E_{12} = [t]_\times \mathrm{R} = \pm U\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}\underbrace{WU^T\mathrm{R}}_{V^T} = \pm U\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}V^T$ with V also orthogonal

this is svd(E) !

$\text{svd}(E_{12}) = \pm U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} W U^T \mathrm{R} = \pm U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$ gives U and V as output matrixes

Two solutions: + sign and – sign

+ sign: → $V^T = W U^T R$ therefore $\mathrm{R}_+ = U W^T V^T = U \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} V^T$

– sign: $-Z = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} W^T$ → $-E_{12} = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} W^T U^T \mathrm{R}$ → $V^T = W^T U^T \mathrm{R}$

→ second solution $\mathrm{R}_- = U W V^T = U \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} V^T$

From $E_{12} = [t]_\times \mathrm{R}$, for each solution for R, there are two solutions for t: $[t]_\times = \pm E_{12} \mathrm{R}^T$

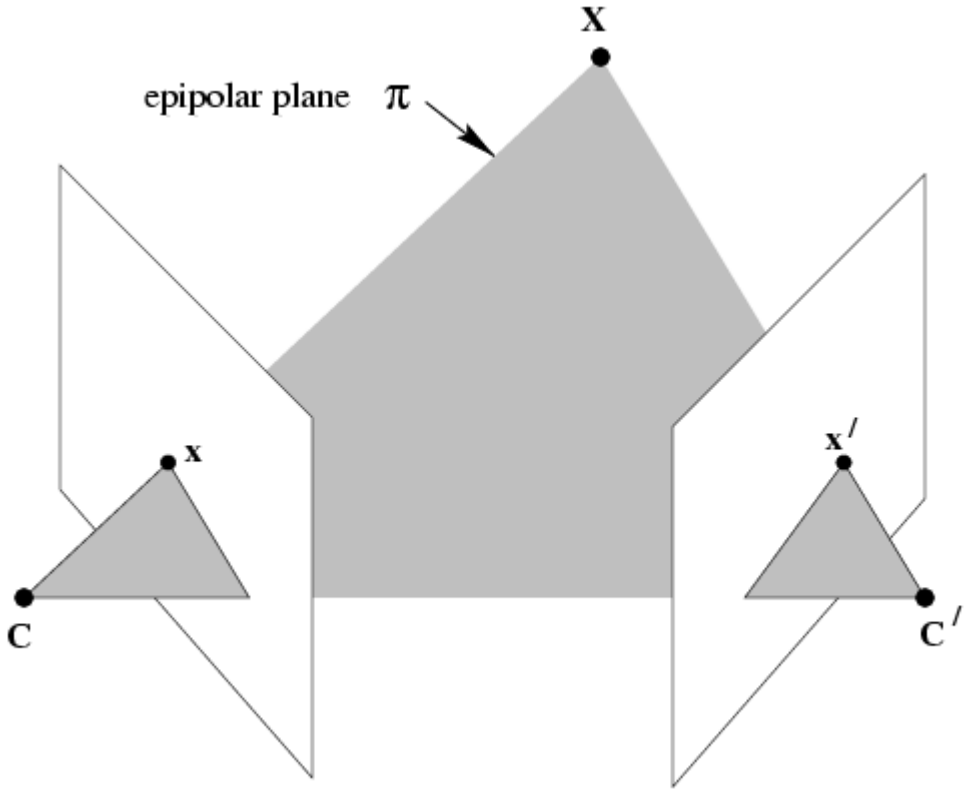
Once R and t have been derived, the relative pose of the two cameras is known, together with $\mathbf{K_1}$ and $\mathbf{K_2}$ → 3D reconstruction by TRIANGULATION

Modulo scale-translation (module of $\boldsymbol{t}$):

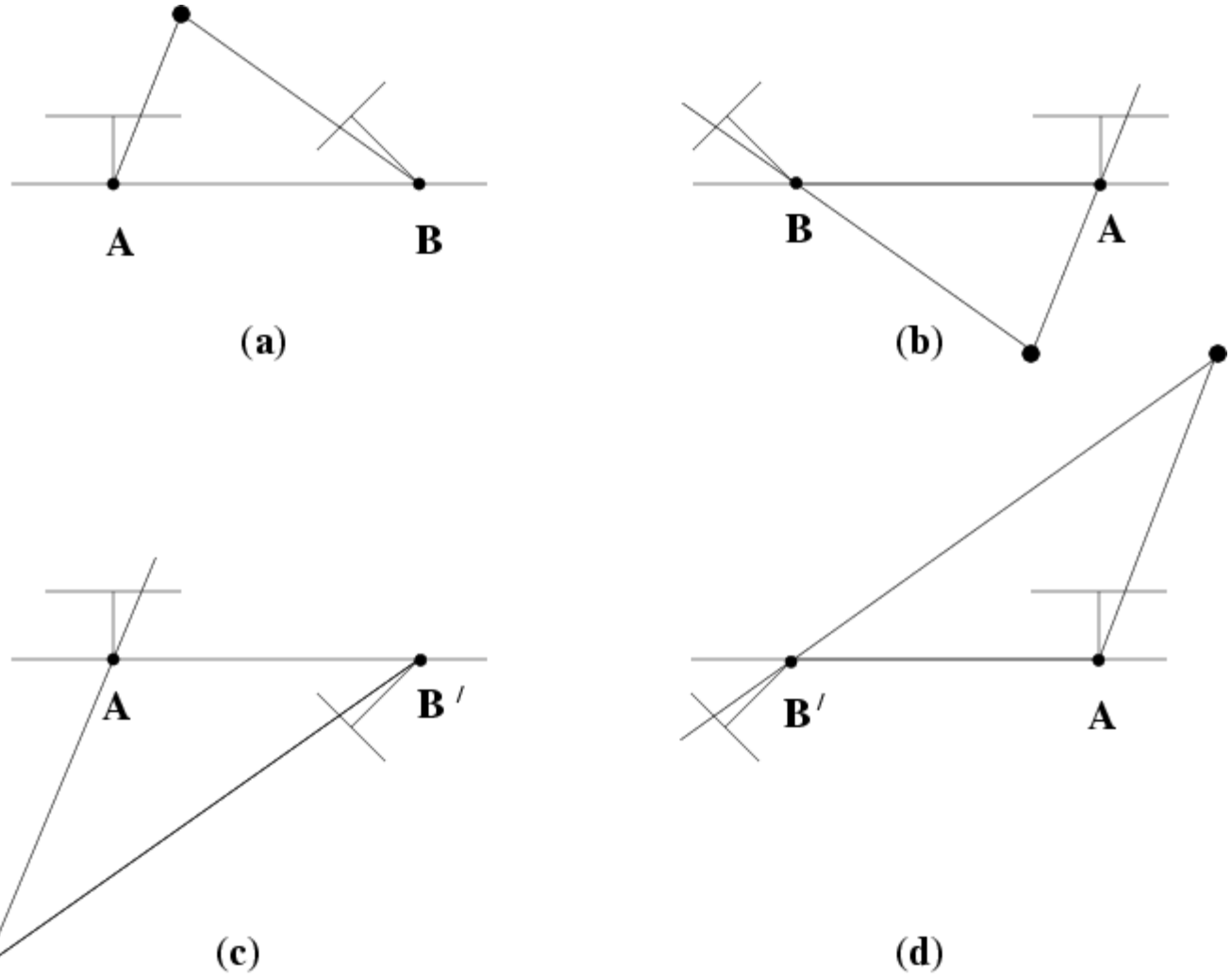
Scale, or module of translation, can not be determined from just images

→Additional information needed, such as,

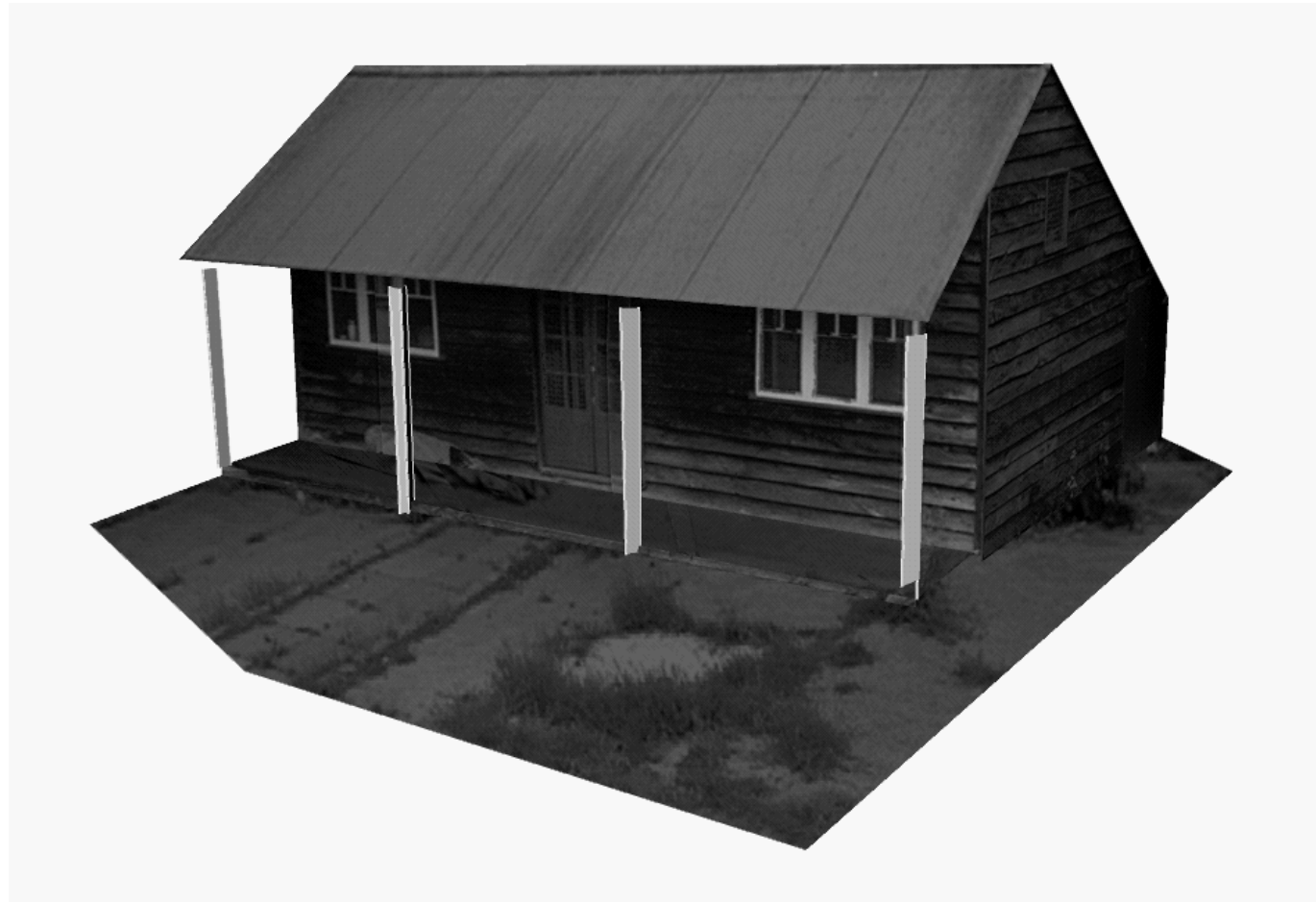
e.g., a size in the scene or $|\boldsymbol{t}|$



Four possible reconstructions from E



(**only one solution** where the triangulated point is **in front** of **both cameras**)



3D reconstruction

from multiple images

Three questions:

(i) **Correspondence geometry:** Given an image point x_1 in the first image, how does this constrain the position of the corresponding point x_2 in the second image?

(ii) **Camera geometry (motion):** Given a set of corresponding image points $\{\mathrm{x}_{1i} \leftrightarrow \mathrm{x}_{2i}\}$, $i = 1, \ldots, n$, what are the cameras P_1 and P_2 for the two views?

(iii) **Scene geometry (structure):** Given corresponding image points $\mathrm{x}_{1i} \leftrightarrow \mathrm{x}_{2i}$ and cameras P_1 , P_2 , what is the position of (their pre-image) X in space?

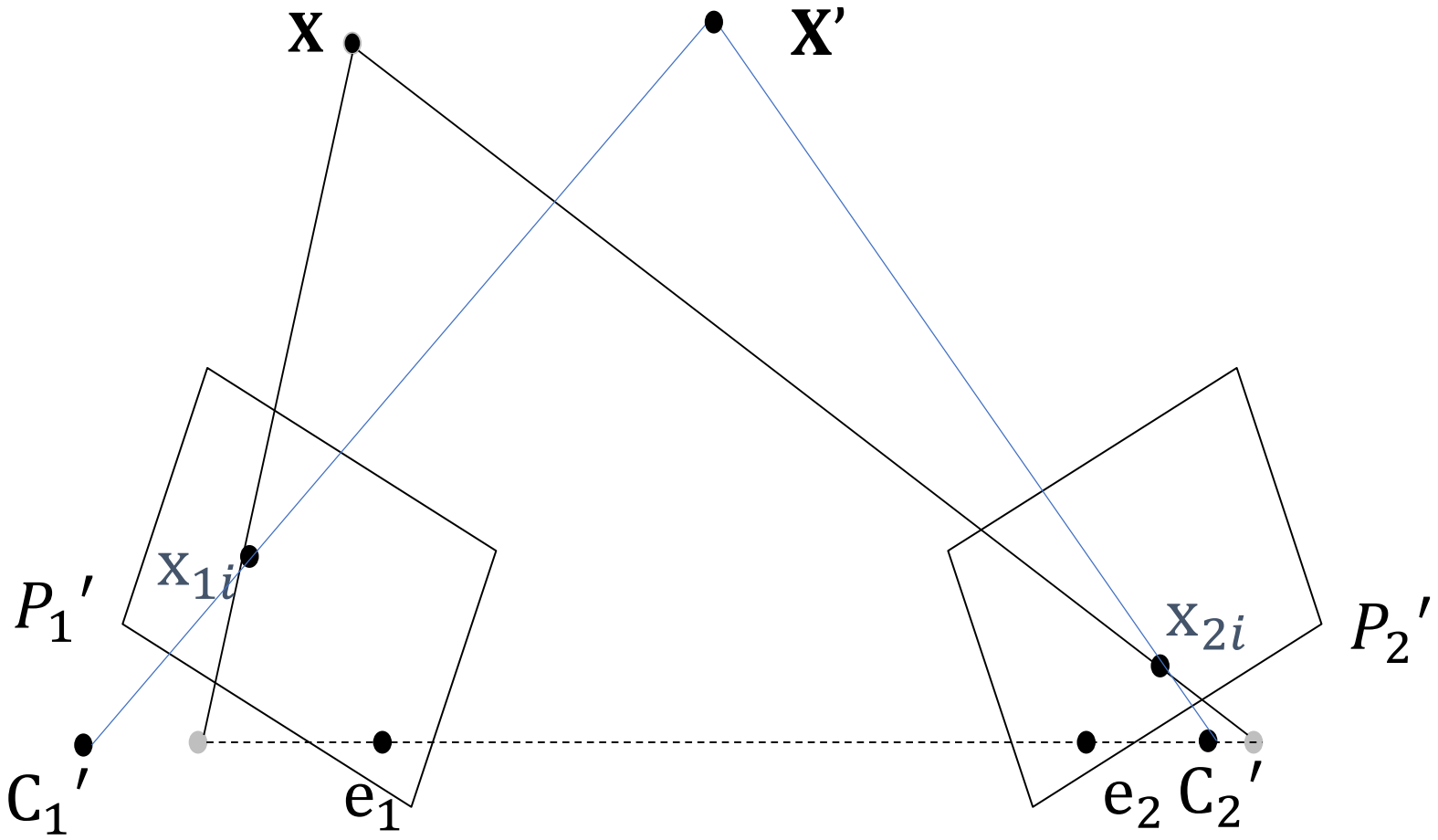
3D reconstruction of cameras and structure

reconstruction problem:

given $\mathrm{x}_{1i} \leftrightarrow \mathrm{x}_{2i}$, compute P_1, P_2 and X_i

$$\mathrm{x}_{1i} = P_1 \mathrm{X}_i \quad \mathrm{x}_{2i} = P_2 \mathrm{X}_i \quad \text{for all } i$$

without additional information, reconstruction is only possible up to projective ambiguity (projective reconstruction)


X
X'
x_{1i}
P_1'
x_{2i}
P_2'
C_1'
e_1
e_2
C_2'

outline of **projective** reconstruction

(i) Compute F from correspondences
(ii) Compute tentative camera matrices from F
(iii) Compute 3D point for each pair of corresponding points

(i) computation of F

use $\mathrm{x}_{2i}^{T}\mathrm{F}\mathrm{x}_{1i} = 0$ equations, linear in coeff. F

8 points (linear), 7 points (non-linear), 8+ (least-squares)

(ii) computation of tentative camera matrices

use $\mathrm{P}_1 = [\mathrm{I}|0] \quad \mathrm{P}_2 = [[\mathrm{e}_2]_\times \mathrm{F}_{12} + \mathrm{e}_2\mathrm{v}^{\mathrm{T}}|\lambda\mathrm{e}_2]$

(iii) triangulation

compute intersection of two backprojected rays

3D projective reconstruction

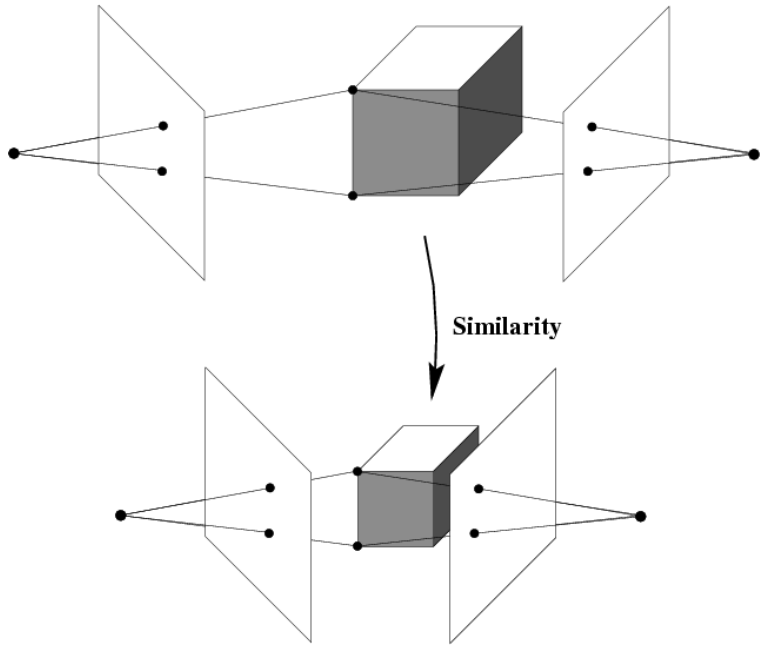
Construct observation of the 3D scene, projective mapping of it

- Take images of 3D scene points X_i and find correspondences $\mathrm{x}_1^i \leftrightarrow \mathrm{x}_2^i$
- Find fundamental matrix(es) F_{12} by solving $\begin{cases} \mathrm{x}_2^{Ti}\mathrm{F}_{12}\mathrm{x}_1^i = 0 \\ i = 1 \ldots N \end{cases}$ for F_{12}
- Take a tentative pair of cameras compatible with F_{12} P_1 P_2

  where e_2 is the LNS of F_{12}, λ and v are any nonzero scalar and vector

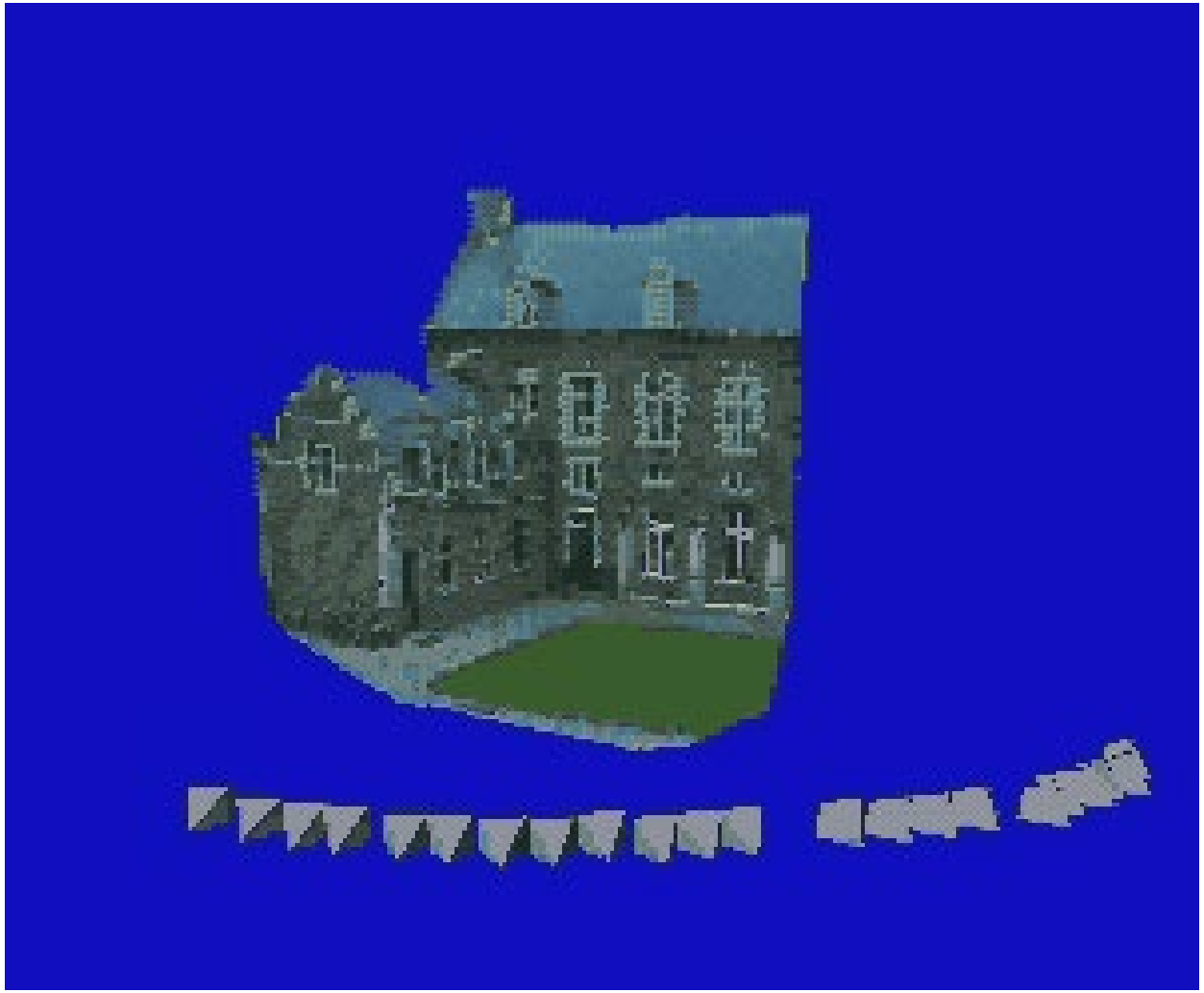
$$\mathrm{P}_1 = [\mathrm{I}|0] \quad \mathrm{P}_2 = [[\mathrm{e}_2]_\times \mathrm{F}_{12} + \mathrm{e}_2\mathrm{v}^\mathrm{T}|\lambda\mathrm{e}_2]$$

Purpose:
reconstruction ambiguity up to a similarity

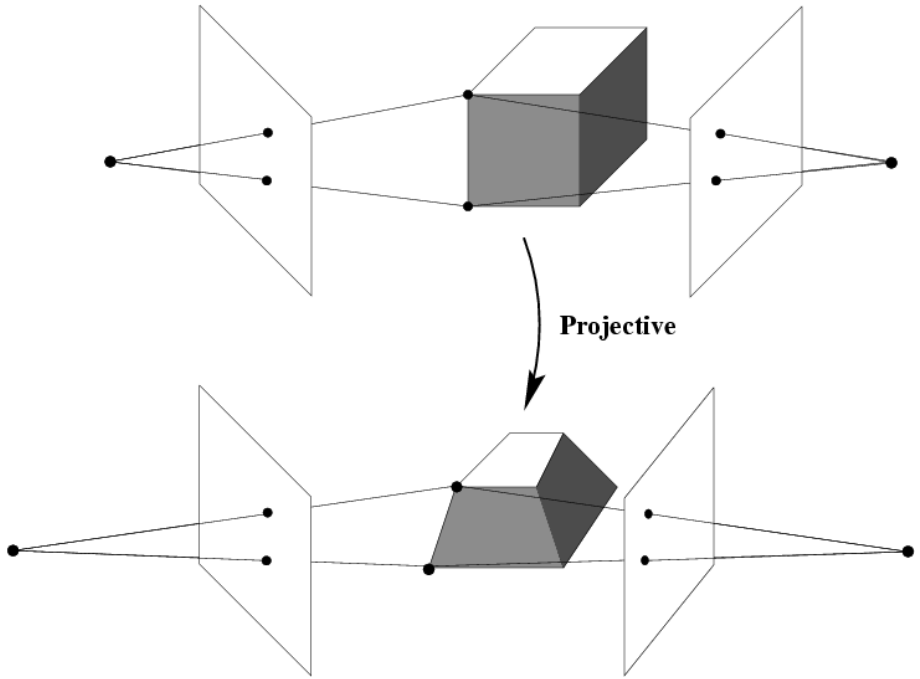


$$x_i = PX_i = \left(PH_S^{-1}\right)\left(H_S X_i\right)$$

$$PH_S^{-1} = K[R|\,t]\begin{bmatrix} R'^T & -R'^T t' \\ 0 & \lambda \end{bmatrix} = K[RR'^T | -RR'^T t' + \lambda t]$$



Starting point:
reconstruction ambiguity up to a projective mapping



$$x_i = PX_i = \left(PH_P^{-1}\right)\left(H_P X_i\right)$$

Terminology

Pairs of corresponding image points $\mathrm{x}_{1i} \leftrightarrow \mathrm{x}_{2i}$

Original unknown scene $\overline{X_i}$

Projective, affine, similarity reconstruction
= reconstruction that is identical to original up to projective, affine, similarity transformation

Literature: Metric and Euclidean reconstruction
= similarity reconstruction

The projective reconstruction theorem

If a set of point correspondences in two views determine the fundamental matrix uniquely, then the scene and cameras may be reconstructed from these correspondences alone, and any two such reconstructions from these correspondences are projectively equivalent

$$x_{1i} \leftrightarrow x_{2i} \qquad (P_1, P_2, \{X_i\}) \qquad (P_1', P_2', \{X_i'\})$$

$$\underbrace{P_1' = P_1 H^{-1} \quad P_2' = P_2 H^{-1}}_{\text{theorem from last class}} \qquad X_i' = H X_i \qquad \left(\begin{matrix}\text{except:} \\ F x_{1i} = x^T{}_{2i} F = 0\end{matrix}\right)$$

$$P_1' X_i' = P_1'(H X_i) = P_1 H^{-1} H X_i = P_1 X_i = x_{1i} = P_1' X_i'$$

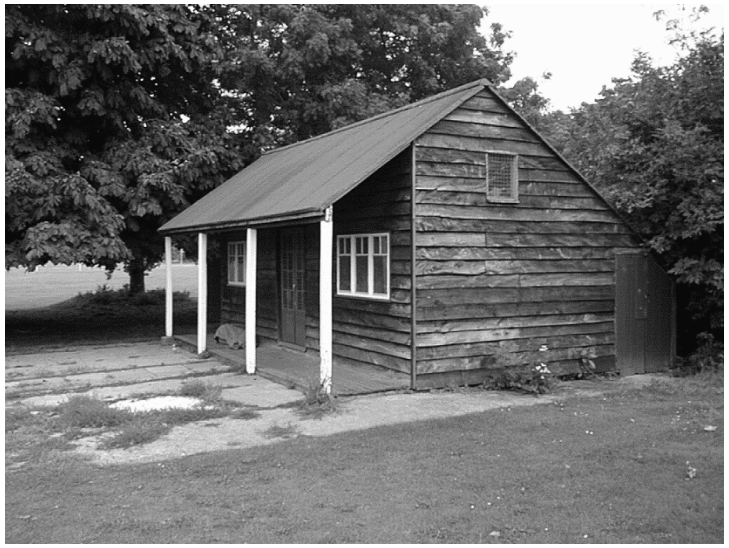
same images idem $P_2 X_i = x_{2i} = P_2' X_i'$

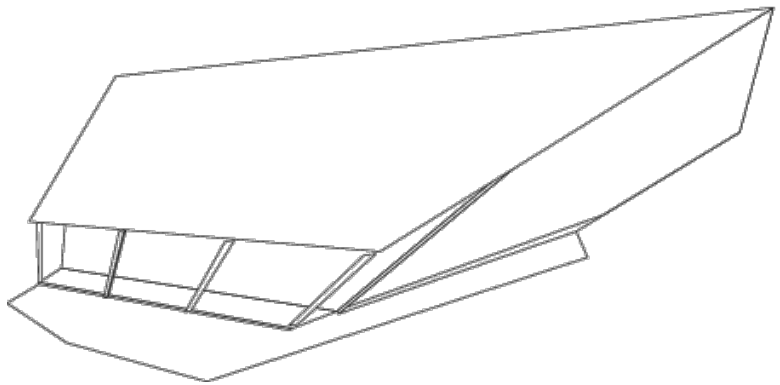
two possibilities: $X_i' = H X_i$, or points along baseline

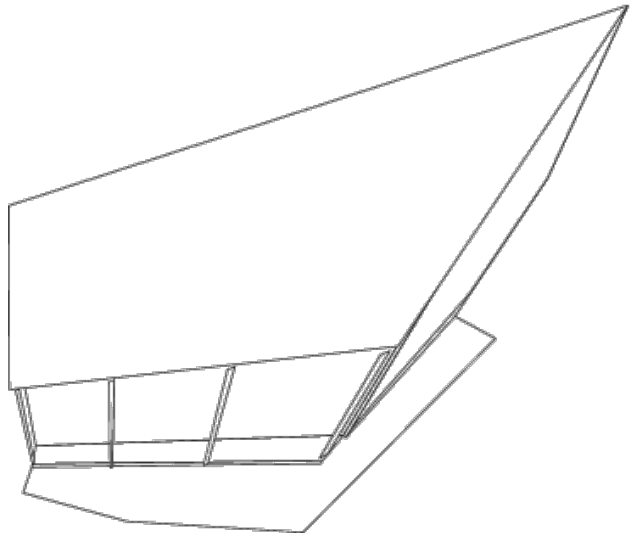
key result:

allows projective reconstruction from pair of uncalibrated images







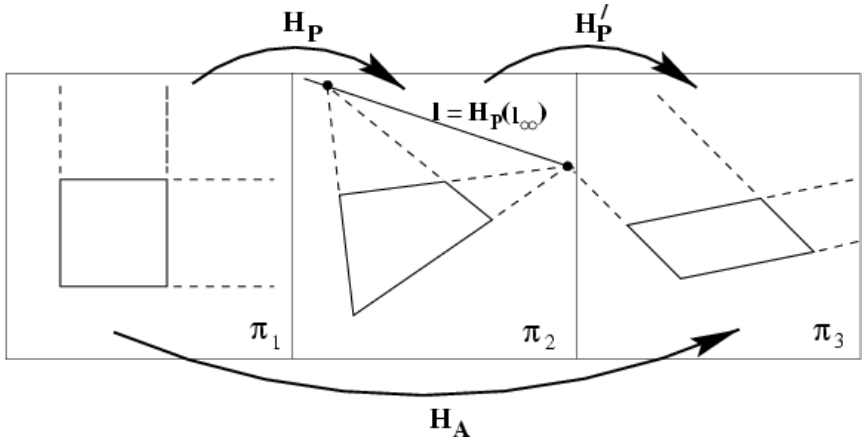


Stratified reconstruction

(i) Projective reconstruction
(ii) Affine reconstruction
(iii) Metric reconstruction

Projective to affine

remember 2-D case



A theorem useful for Affine reconstruction

Theorem on an affine invariant

Theorem. *A projective transformation H maps the plane at the infinity $\boldsymbol{\pi_\infty}$ onto itself, i.e., $\boldsymbol{\pi_\infty}$ is **invariant** under a projective transformation*

$\updownarrow$

H *is* ***affine***

Application to affine reconstruction

Given 3D points obtained by an uknown projective mapping of an unknown original scene (set of points in 3D space)

→

the plane $\boldsymbol{\pi}'_{\infty}$ (i.e. the transformed $\boldsymbol{\pi}_{\infty}$) is in general $\neq \boldsymbol{\pi}_{\infty}$!!

Use $\boldsymbol{\pi}'_{\infty}$ as additional information: if we apply to the transfomed set a second mapping H_{AR} which maps $\boldsymbol{\pi}'_{\infty}$ back to $\boldsymbol{\pi}_{\infty}$, we obtain a new, reconstructed model

The composed mapping maps $\boldsymbol{\pi}_{\infty}$ again to $\boldsymbol{\pi}_{\infty}$ →

From the theorem, the obtained model is an affine mapping of the original scene

→

The obtained model is an **affine reconstruction** of the scene

Use of $\boldsymbol{\pi'_\infty}$ in affine reconstruction

….

apply to the transformed point set a second projective mapping $\boldsymbol{H_{AR}}$

that maps $\boldsymbol{\pi'_\infty}$ back to $\boldsymbol{\pi_\infty}$,

how can we find such a projective mapping H_{AR} ?

$$H_{AR} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ & \boldsymbol{\pi'}_\infty^{T} & & \end{bmatrix},$$

such that H_{AR} it is invertible

To sum up: affine rectification from $\boldsymbol{\pi'_\infty}$

- Find three points $\mathbf{X_1}, \mathbf{X_2}, \mathbf{X_3}$ that result from mapping three points **at the infinity**
- Fit the transformed plane $\boldsymbol{\pi'_\infty}$ to them: $\boldsymbol{\pi'_\infty} = \mathbf{RNS}\left(\begin{bmatrix} \mathbf{X_1}^T \\ \mathbf{X_2}^T \\ \mathbf{X_3}^T \end{bmatrix}\right)$
- Affine rectification matrix

$$H_{AR} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ & \boldsymbol{\pi'_\infty}^T & & \end{bmatrix}$$

- Affine reconstructed model $M_A = \{\mathbf{X'}_i\} = H_{AR}\ \text{given}_{\text{points}} = H_{AR}\ \{\mathbf{X}_i\}$

Theorem
A projective transformation $\mathcal{H}$ maps π_∞ to itself

$\updownarrow$

$\mathcal{H}$ is an affinity

if π'_∞ known → apply mapping from π'_∞ to π_∞:
combined mapping from real to reconstructed is then an affinity

Projective to affine: additional information π'_∞

$(P_1, P_2, \{X_i\})$ initial cameras, e.g., canonical form

$$\pi'_\infty = (A, B, C, D)^{\mathrm{T}} \mapsto (0,0,0,1)^{\mathrm{T}} = \pi_\infty$$

$$\mathrm{H}^{-\mathrm{T}}\pi'_\infty = (0,0,0,1)^{\mathrm{T}}$$

$$\mathbf{H} = \begin{bmatrix} \mathbf{I}|\mathbf{0} \\ \boldsymbol{\pi}'_\infty \end{bmatrix} \qquad \text{(if } D \neq 0\text{)}$$

theorem says up to a projective transformation,
but a projective transformation with fixed π_∞ is **affine**

new cameras: $P'_1 = P_1 H^{-1}$ and $P'_2 = P_2 H^{-1}$
new reconstruction: $X'_i = H\ X_i$ affine mapping of the true points

Affine reconstruction can be sufficient for some applications,
e.g. mid-point, centroid, parallellism

constraints on π'_∞: examples

Projective to affine

Costraints on π'_∞ from translational motion

points at infinity are fixed for a pure translation

$\Rightarrow$ Reconstruction (triangulation) of $x_i \leftrightarrow x_i$ is on π_∞



$$F = [e']_\times M M^{-1} = [e']_\times = [e]_\times$$

e': vanishing point of motion direction

$$P = [I|0]$$

$$P' = [I|e']$$

Constraints on π'_∞ from scene

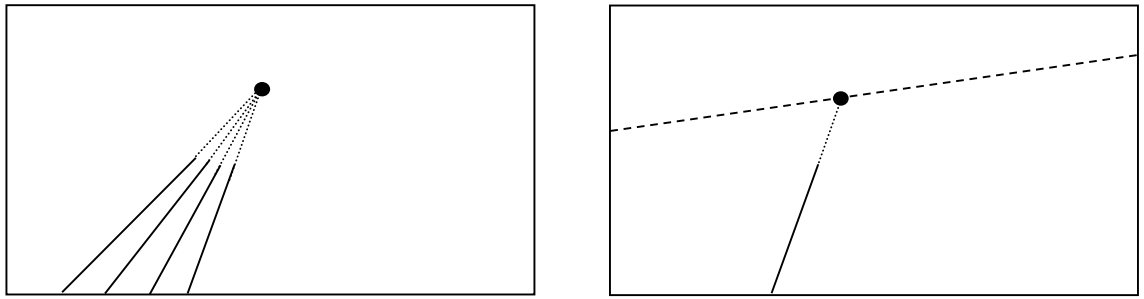
Parallel lines observed in the images

parallel lines intersect at infinity
reconstruction of corresponding vanishing point yields
a point whose **true position** is on the **plane at infinity**

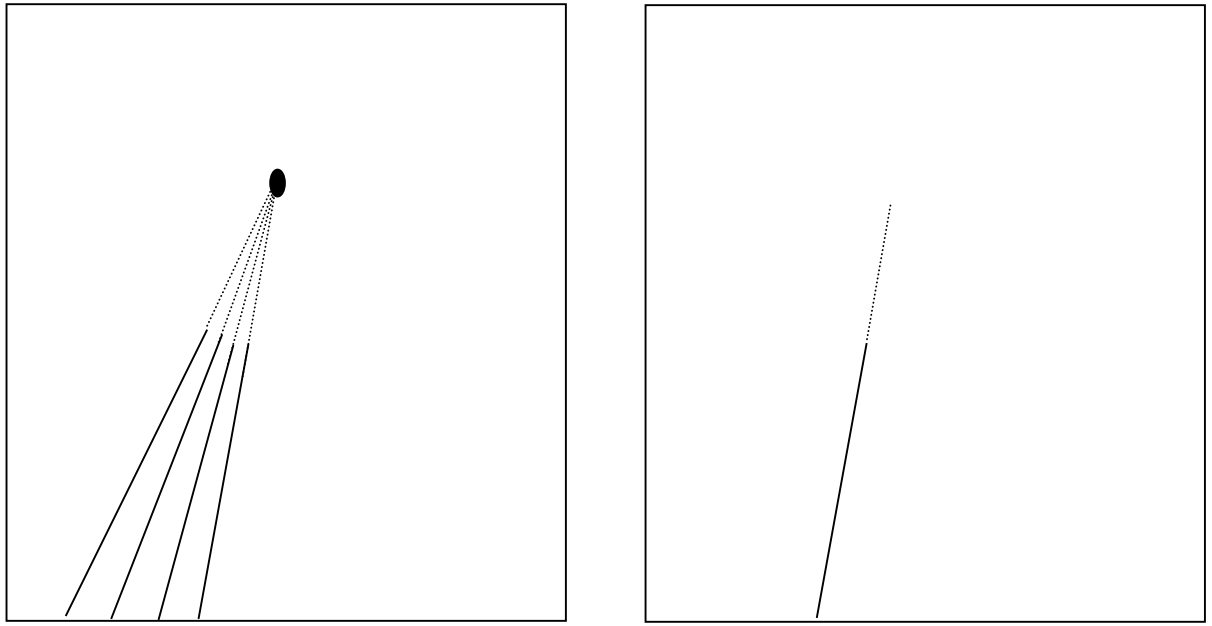
Images of 3 sets of parallel lines allow to uniquely determine π'_∞

remark: in presence of noise determining the intersection of parallel lines is a delicate problem

remark: obtaining vanishing point in one image, and just one line in the other image, can be sufficient

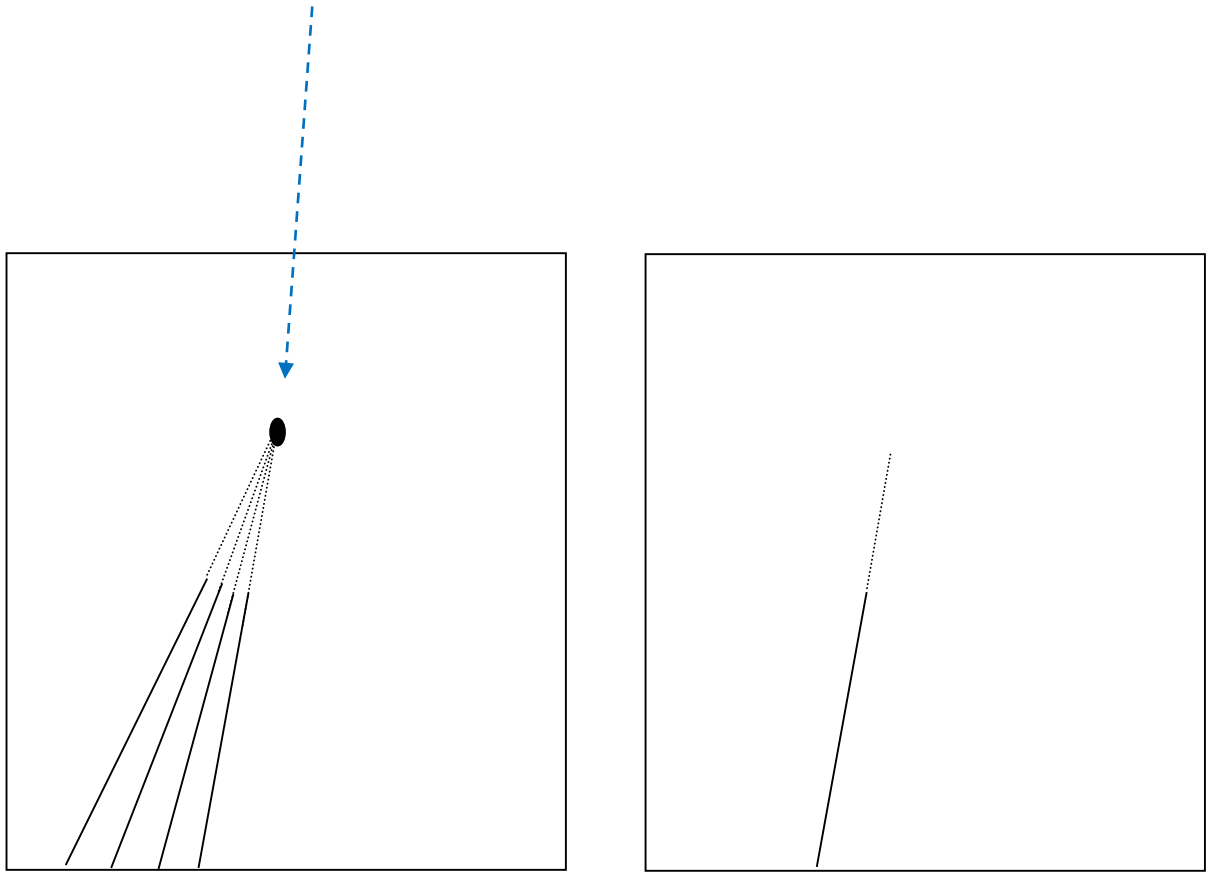


Projective to affine



remark: obtaining vanishing point in one image, and just one line in the other image, can be sufficient

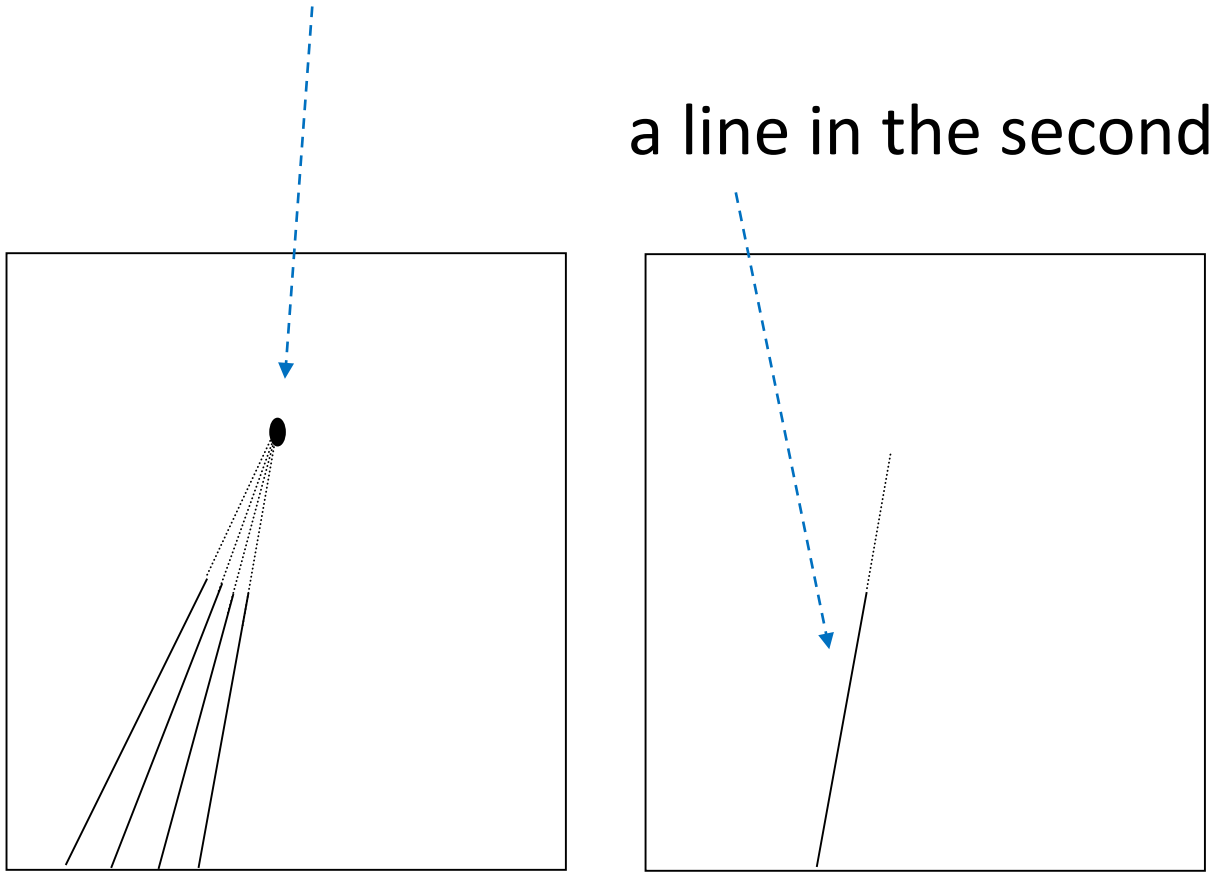
vanishing point in the first image



remark: obtaining vanishing point in one image, and just one line in the other image, can be sufficient

vanishing point in the first image

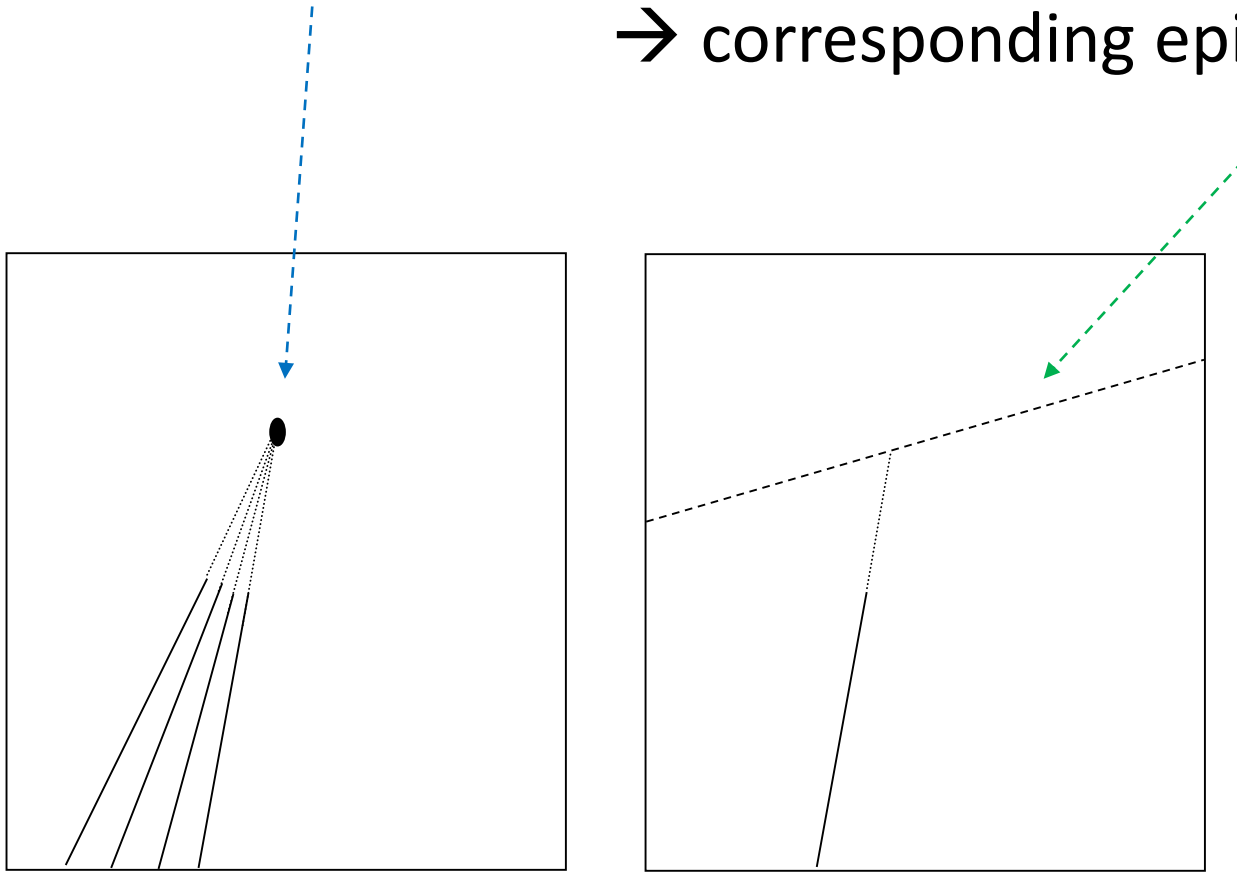
a line in the second image



remark: obtaining vanishing point in one image, and just one line in the other image, can be sufficient

vanishing point V in the first image

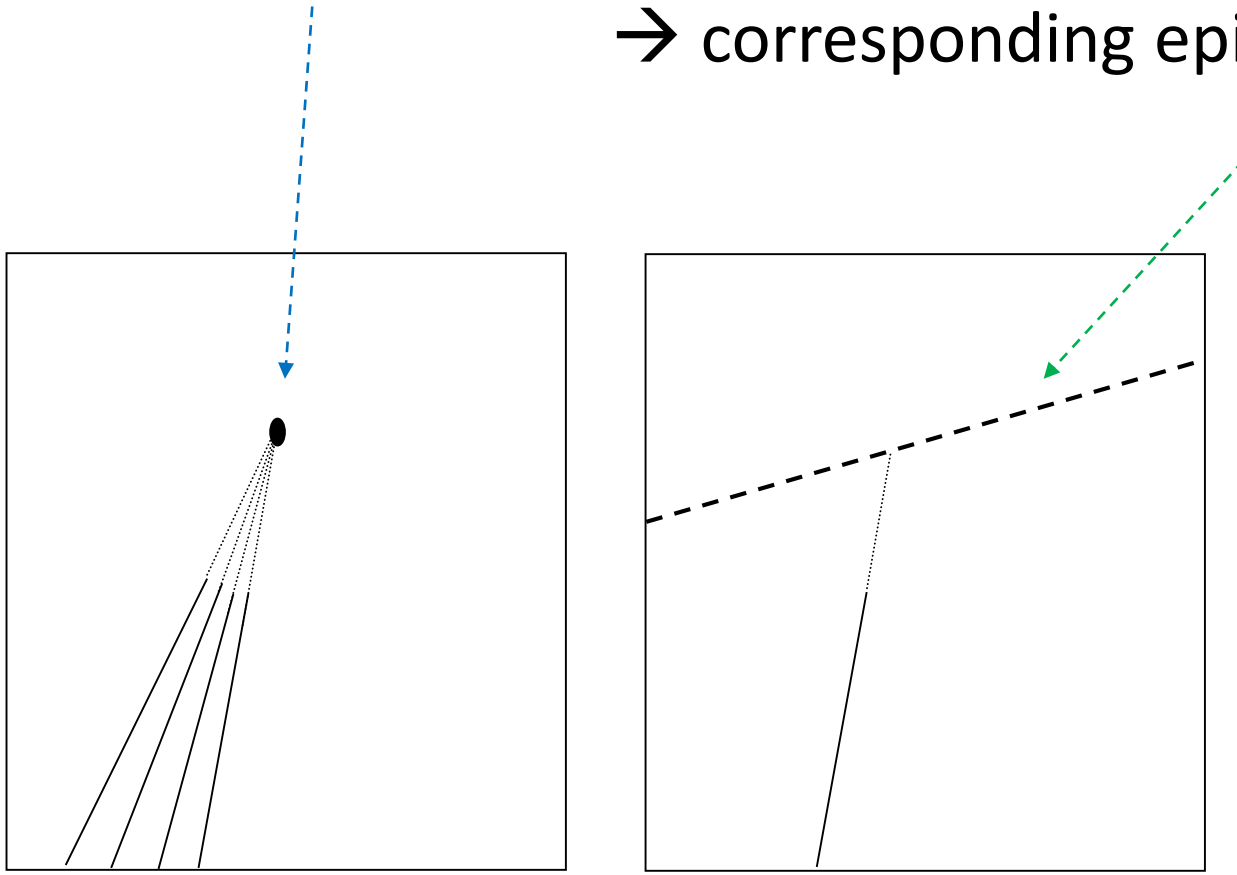
→ corresponding epipolar line l’ = FV



remark: obtaining vanishing point in one image, and just one line in the other image can be sufficient

vanishing point V in the first image

→ corresponding epipolar line l’ = FV

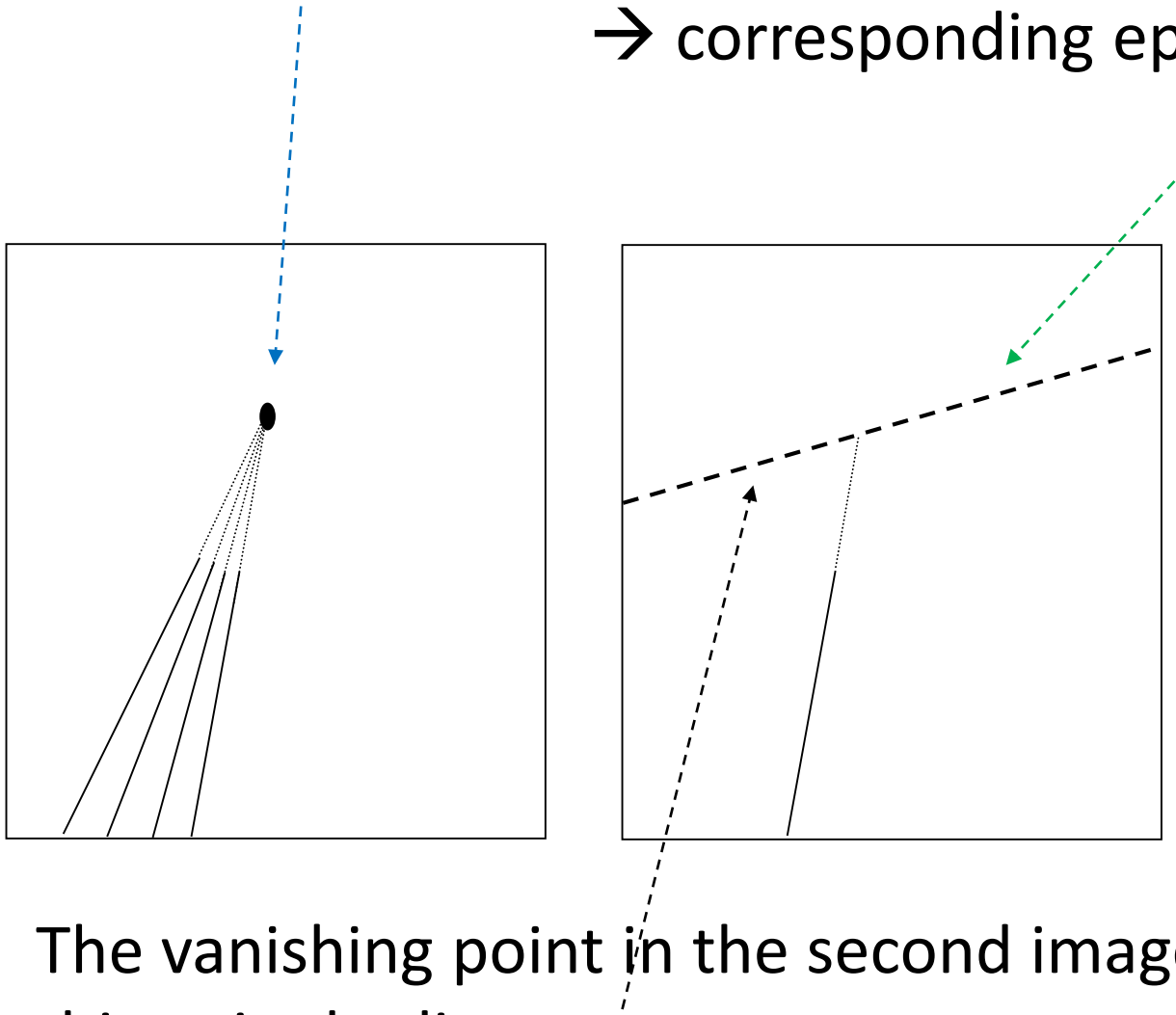


The vanishing point in the second image …

remark: obtaining vanishing point in one image, and just one line in the other image can be sufficient

vanishing point V in the first image

→ corresponding epipolar line l’ = FV

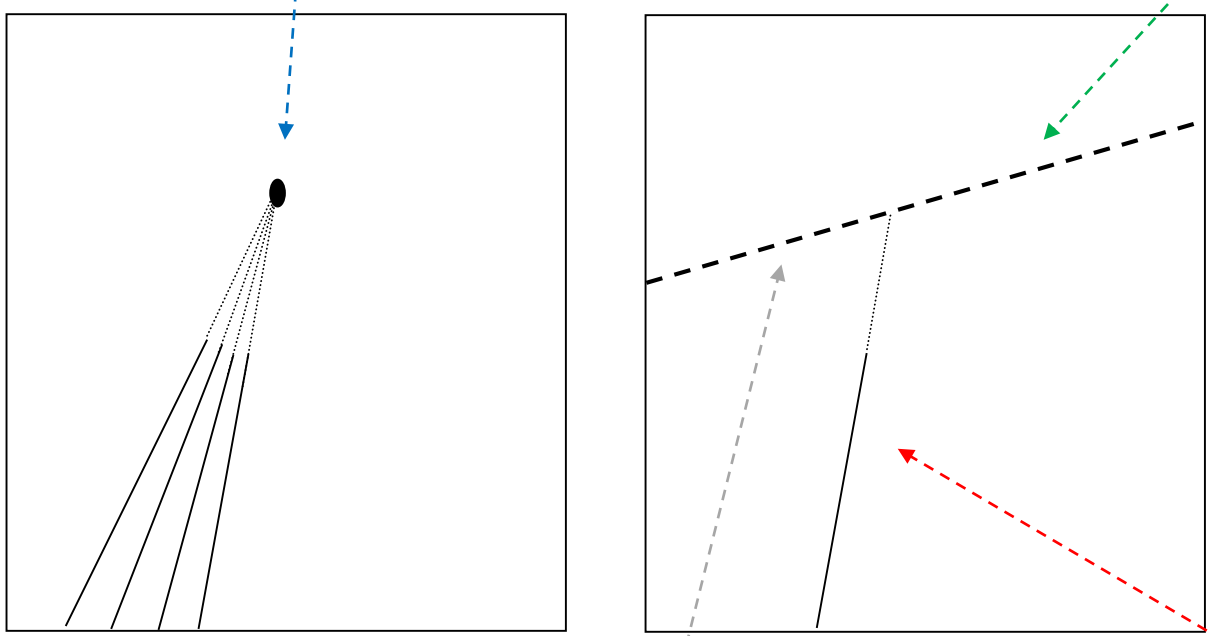


The vanishing point in the second image must lie (i) on this epipolar line

remark: obtaining vanishing point in one image, and just one line in the other image can be sufficient

vanishing point V in the first image

→ corresponding epipolar line l’ = FV

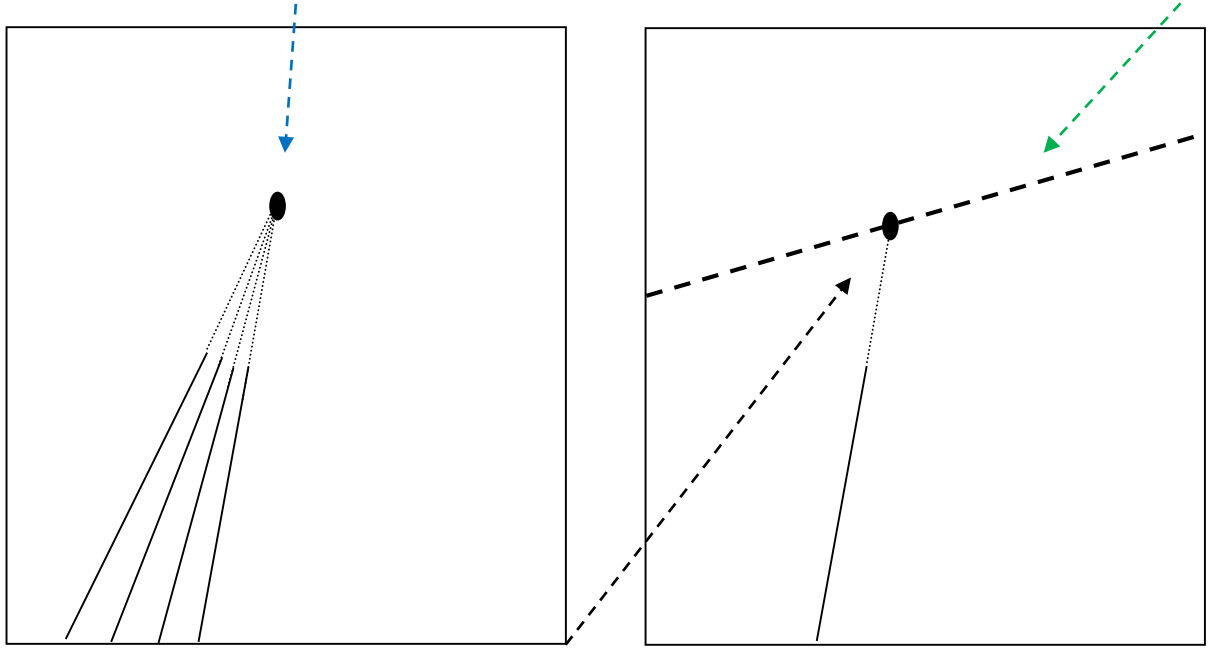


The vanishing point in the second image must lie (i) on this epipolar line, and (ii) on this image line

remark: obtaining vanishing point in one image, and just a line in the other image can be sufficient

vanishing point V in the first image

→ corresponding epipolar line l’ = FV

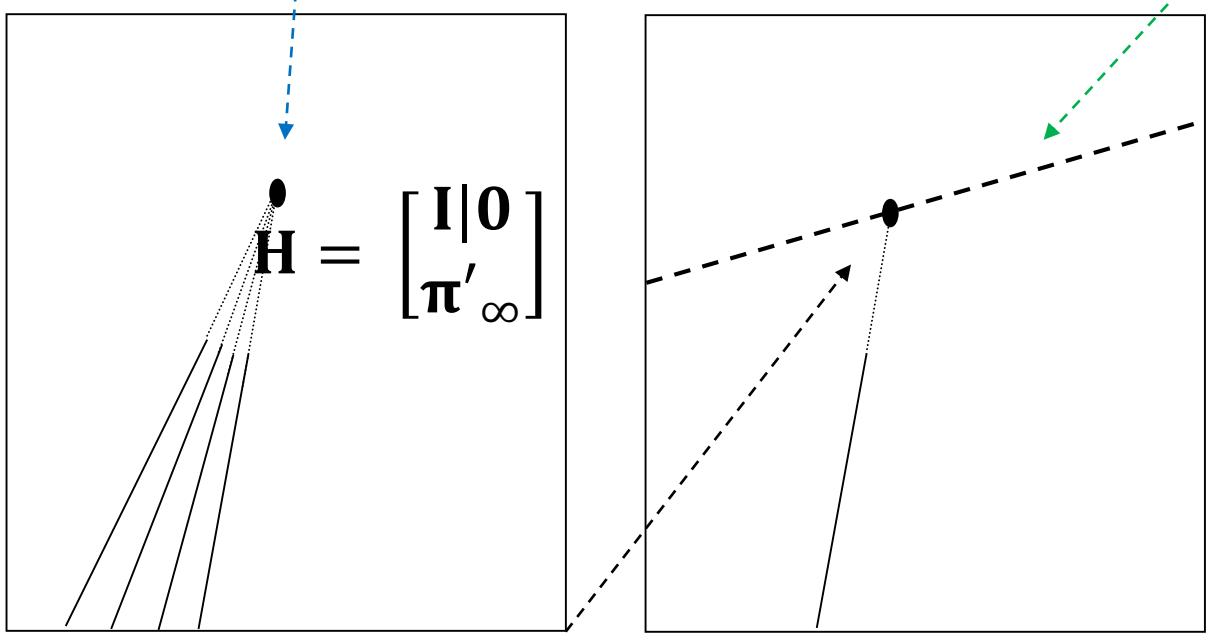


The vanishing point in the second image must lie (i) on this epipolar line, and (ii) on this image line

remark: obtaining vanishing point in one image, and just a line in the other image can be sufficient

vanishing point V in the first image

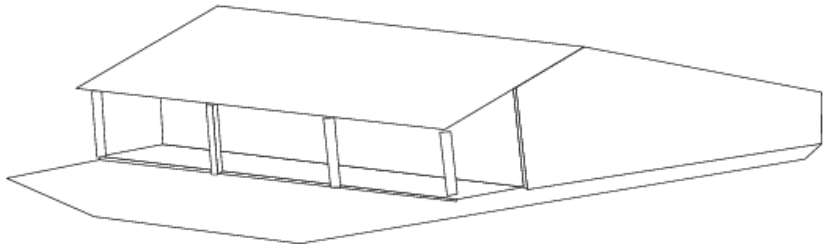
→ corresponding epipolar line l’ = FV

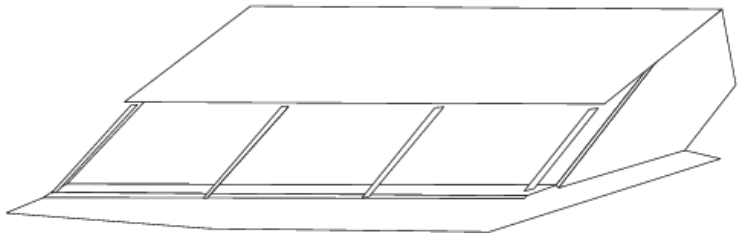


From 3 such pairs of corresponding vanishing points triangulate them → find 3 3D points → find π'_{∞} through them → apply $\mathbf{H} = \begin{bmatrix} \mathbf{I}|\mathbf{0} \\ \boldsymbol{\pi}'_{\infty} \end{bmatrix}$

Scene constraints: parallel lines





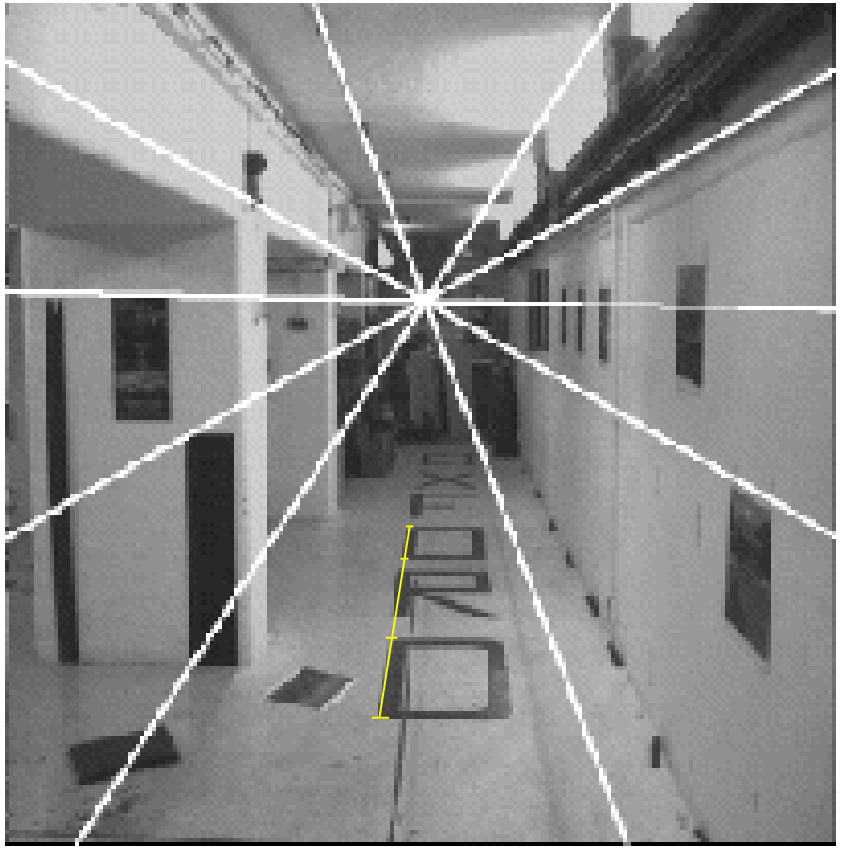


Projective to affine

other constraints on π'_∞ from scene

known distance ratios on a line

known distance ratio along a line allow to determine point at infinity (same as 2D case) : from cross ratio

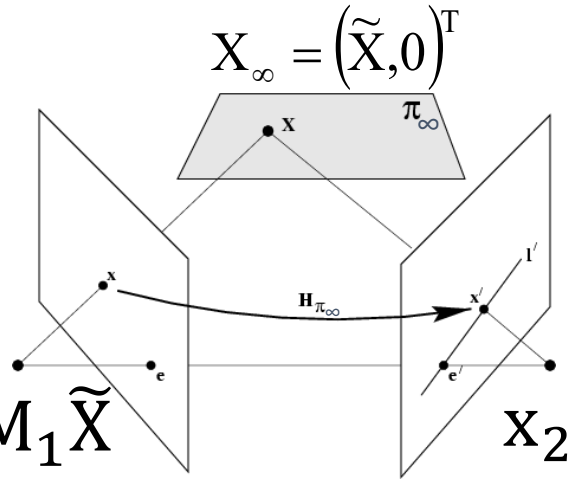


Projective to affine

Alternative cameras
(for affine reconstruction)

The homography induced by the plane at the ∞

true cameras $P_1 = [M_1|\mathbf{m}_1] \quad P_2 = [M_2|\mathbf{m}_2]$



$$x_1 = M_1\tilde{X} \qquad x_2 = M_2\tilde{X}$$

$$x_2 = M_2\tilde{X} = M_2 {M_1}^{-1} x_1$$

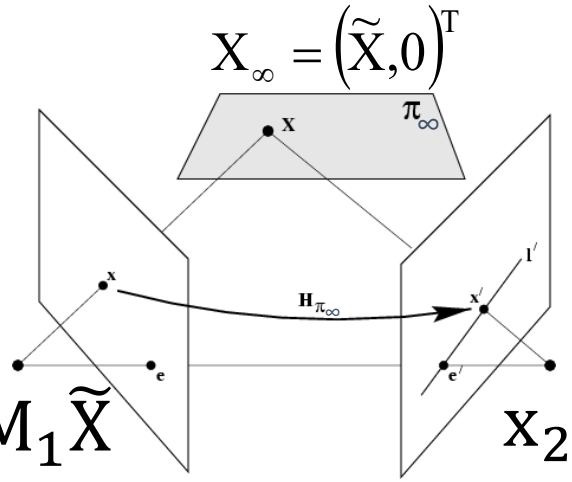
$$H_\infty = M_2 {M_1}^{-1}$$

Projective to affine

The homography induced by the plane at the ∞

true cameras $P_1 = [M_1|\mathbf{m}_1] \quad P_2 = [M_2|\mathbf{m}_2]$

$$H_\infty = M_2 {M_1}^{-1}$$



$$x_1 = M_1 \tilde{X} \qquad x_2 = M_2 \tilde{X}$$

but also … unchanged under **affine transformations**

$$P_i = [M_i|m_i] \begin{bmatrix} A & a \\ 0 & 1 \end{bmatrix} = [M_i A | M_i a + m_i]$$

$$\rightarrow H_\infty = M_2 A A^{-1} {M_1}^{-1} = M_2 {M_1}^{-1}$$

affine reconstruction: alternative cameras $P_1 = [I|0] \quad P_2 = [H_\infty | e]$

Proof: use affine transformation with H^{-1} where $H \stackrel{\text{def}}{=} \begin{bmatrix} A & a \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} M & m \\ 0 & 1 \end{bmatrix}$

Projective to affine

The homography induced by the plane at the ∞

true cameras $\mathrm{P}_1 = [\mathrm{M}_1|\mathbf{m}_1] \quad \mathrm{P}_2 = [\mathrm{M}_2|\mathbf{m}_2]$ $\qquad \mathrm{H}_\infty = \mathrm{M}_2\mathrm{M}_1^{-1}$

and also … affine reconstruction: alternative cameras $\mathrm{P'}_1 = [\mathrm{I}|0] \quad \mathrm{P'}_2 = [\mathrm{H}_\infty|\mathrm{e}]$

Proof: use affine transformation with H^{-1} where $\mathrm{H} \stackrel{\text{def}}{=} \begin{bmatrix} \mathrm{M}_1 & \mathbf{m}_1 \\ 0 & 1 \end{bmatrix}$ →

$$\mathrm{H}^{-1} = \begin{bmatrix} \mathrm{M}_1^{-1} & -\mathrm{M}_1^{-1}\mathbf{m}_1 \\ 0 & 1 \end{bmatrix}$$

→ $\mathrm{P'}_1 = \mathrm{P}_1\mathrm{H}^{-1} = [\mathrm{I}|0]$

and

→ $\mathrm{P'}_2 = \mathrm{P}_2\mathrm{H}^{-1} = \left[\mathrm{M}_2\mathrm{M}_1^{-1}\middle| -\mathrm{M}_2\mathrm{M}_1^{-1}\mathbf{m}_1 + \mathbf{m}_2\right] = [\mathrm{H}_\infty|\mathrm{e}]$

in fact the first camera center is $\mathbf{o}_1 = -\mathrm{M}_1^{-1}\mathbf{m}_1$

and its image is

$$\mathrm{e} = \mathrm{P}_2\begin{bmatrix} \mathbf{o}_1 \\ 1 \end{bmatrix} = [\mathrm{M}_2|\mathbf{m}_2]\begin{bmatrix} \mathbf{o}_1 \\ 1 \end{bmatrix} = -\mathrm{M}_2\mathrm{M}_1^{-1}\mathbf{m}_1 + \mathbf{m}_2$$

Alternative cameras
(for affine reconstruction)

Alternative cameras
(for affine reconstruction)

Remember:

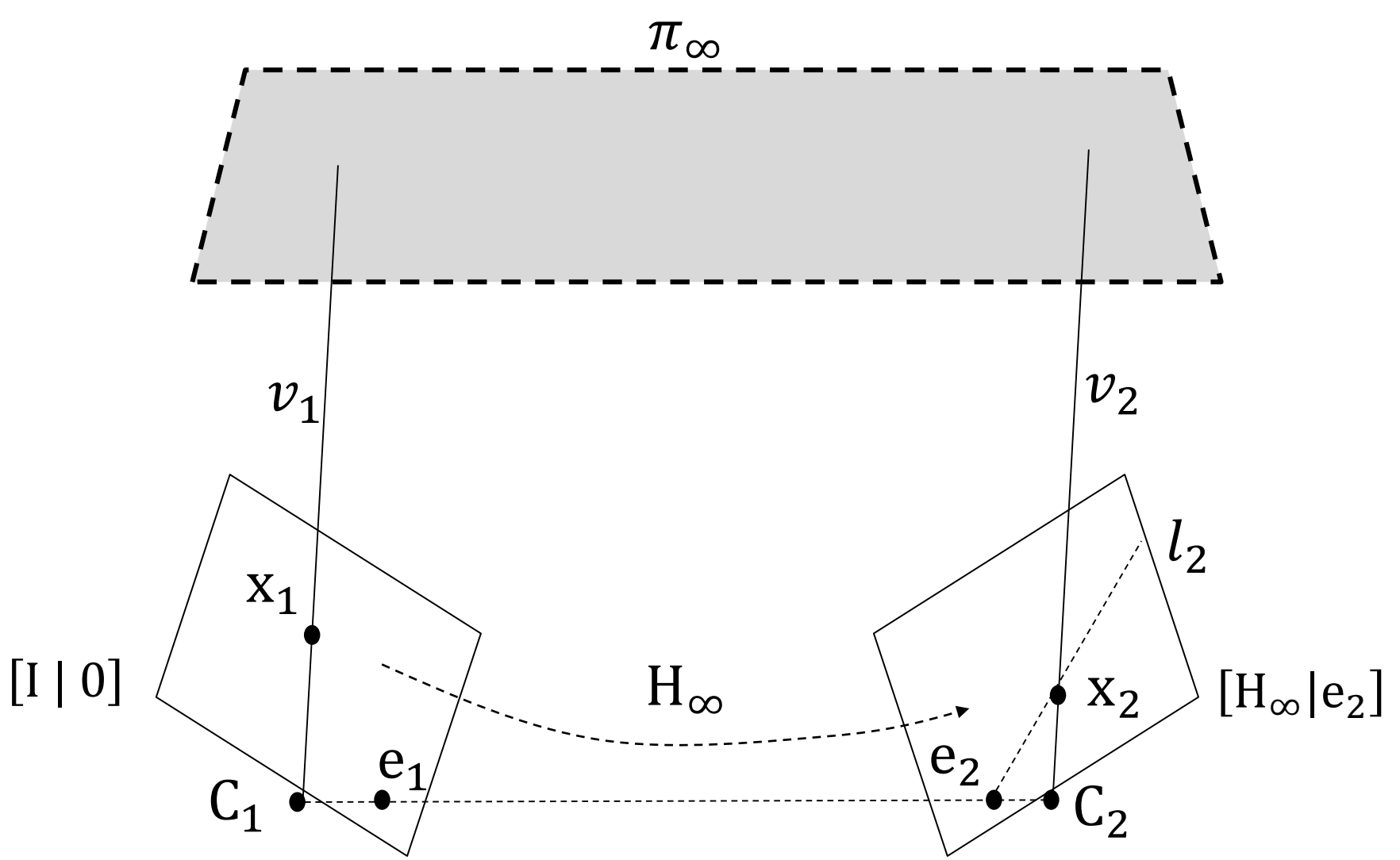
if 3D points are on a plane π, their two images are related by a homography H_{π} induced by plane π

Alternative cameras (for affine reconstruction)

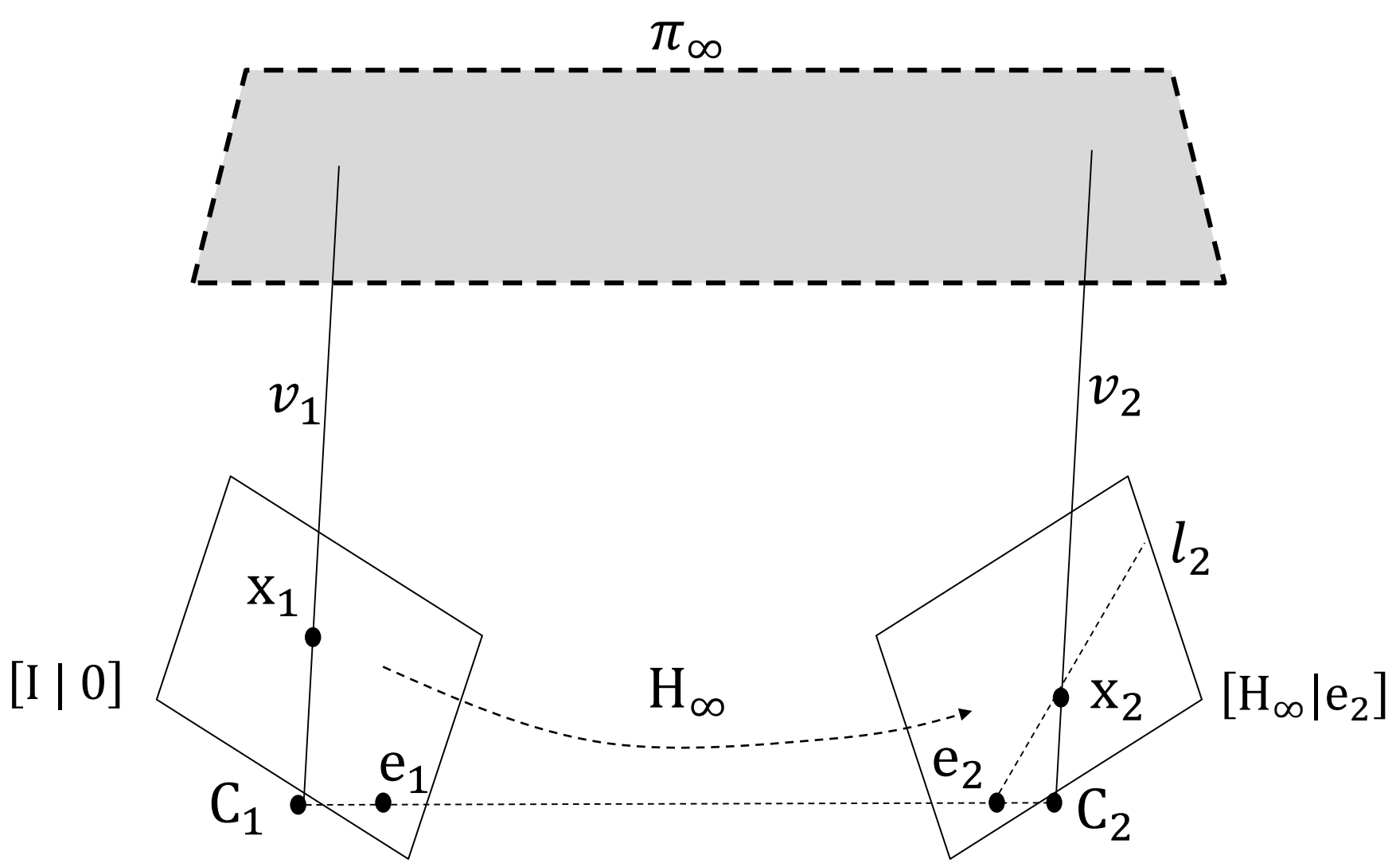
if 3D points are known to be at the ∞, i.e., on the plane π_∞, their two images are related by the homography H_∞ induced by plane π_∞

This homography H_∞ can be computed by three correspondences $\mathrm{x}_1 \leftrightarrow \mathrm{x}_2$ + one $\mathrm{e}_1 \leftrightarrow \mathrm{e}_2$

Now check camera pair $([\mathrm{I} \mid 0]\ ,\ [\mathrm{H}_\infty | \mathrm{e}_2])$



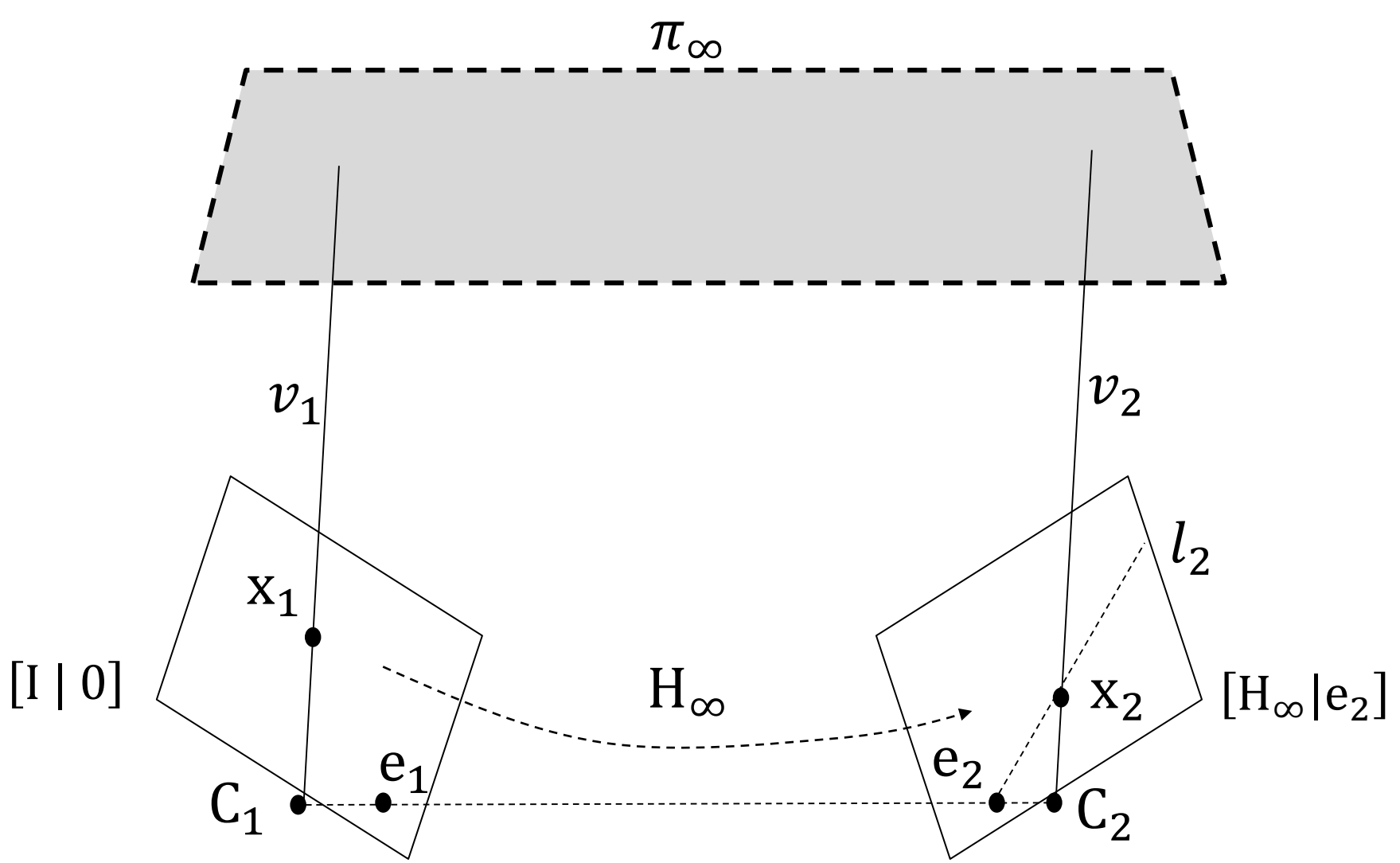
triangulate vewing rays of $x_1 \leftrightarrow x_2 = H_\infty x_1$ using cameras $[I \mid 0]$: dir(v_1) $= I^{-1}x_1 = x_1$ and …



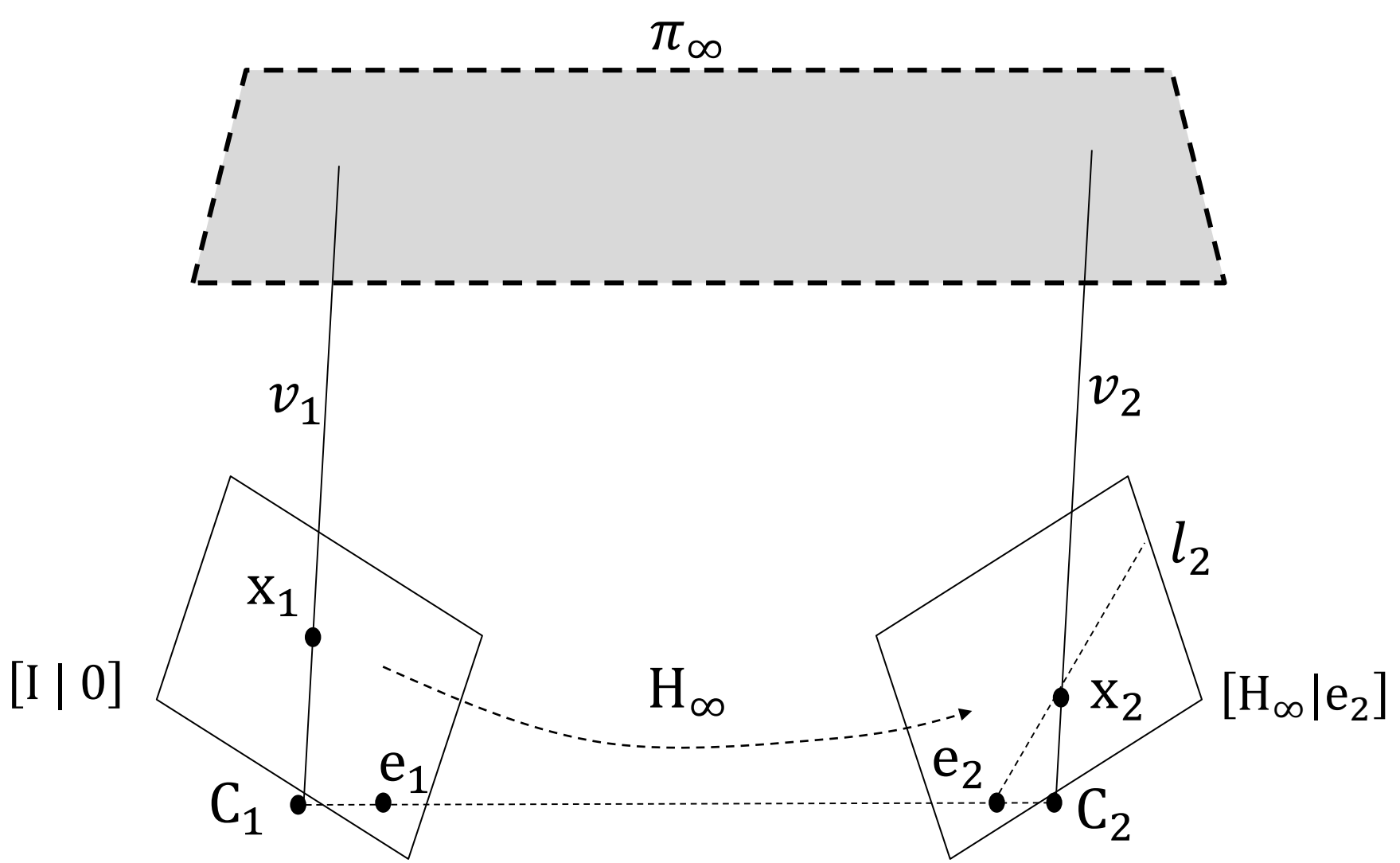
triangulate vewing rays of $x_1 \leftrightarrow x_2 = H_\infty x_1$ using cameras

$[I \mid 0]$: dir$(v_1) = I^{-1}x_1 = x_1$ and

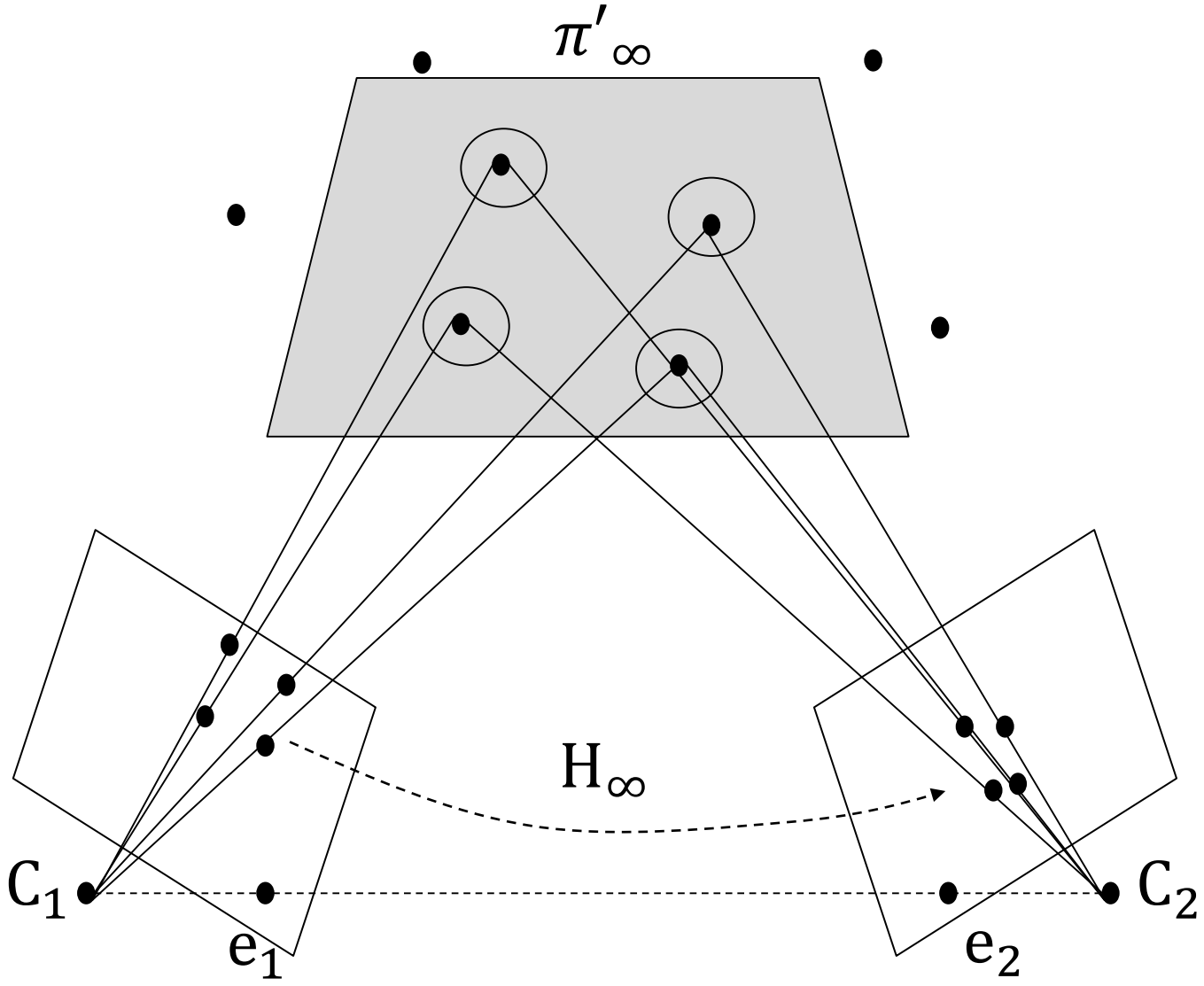
$[H_\infty|e_2]$: dir$(v_2) = {H_\infty}^{-1}x_2 = {H_\infty}^{-1}H_\infty x_1 = x_1 =$ dir(v_1)



triangulate vewing rays of $x_1 \leftrightarrow x_2 = H_\infty x_1$ using cameras $[I \mid 0]$: dir$(v_1) = x_1$ and $[H_\infty|e_2]$: dir$(v_2) = x_1 =$ dir(v_1)
viewing rays v_1 , v_2 are parallel → their ∩ X' is at the ∞ !!



true point X on π_∞, reconstructed point X' on π_∞: AFFINE RECONSTRUCTION



Affine reconstruction: from 4 pairs of corresponding **vanishing** points compute homography H_π (H_∞). Use cameras $[I \mid 0]$ and $[H_\infty | e_2]$ to triangulate 3D points

To compute the homography induced by the plane at the infinity

- find three pairs of corresponding vanishing points $\mathrm{v}_{1\mathrm{j}} \leftrightarrow \mathrm{v}_{2\mathrm{j}}$
  $j = 1..3$ from two images

- use epipoles $\mathrm{e}_1 \leftrightarrow \mathrm{e}_2$ as the fourth pair of points (the epipoles can not be used to triangulate a point in 3D)

- → compute homography H_∞ using these four pairs of points

then use cameras $\mathrm{P}_1 = [\mathrm{I}|0]$ and $\mathrm{P}_2 = [\mathrm{H}_\infty|\mathrm{e}_2]$ as new cameras

affine to metric

Affine to metric

Given $\mathrm{P}_i = [\mathrm{M}_i|\mathrm{m}_i]$ and $\omega_\mathrm{i} = \left(\mathrm{K}\mathrm{K}^\mathrm{T}\right)^{-1}$ of **just one** of the cameras, a possible transformation from affine to metric is

$$\mathrm{H} = \begin{bmatrix} \mathrm{A}^{-1} & 0 \\ 0 & 1 \end{bmatrix} \text{ with } \mathrm{A} = \text{Cholesky factorisation of } \quad \mathrm{A}\mathrm{A}^\mathrm{T} = \left(\mathrm{M}_i{}^\mathrm{T}\omega_\mathrm{i}\mathrm{M}_i\right)^{-1}$$

Proof: new camera $\mathrm{P}_i' = \mathrm{P}_i\,\mathrm{H}^{-1} = [\mathrm{M}_i\mathrm{A}|\mathrm{m}_i]$ is OK if

$$\rightarrow$$

$$\omega_\mathrm{i}{}^{-1} = \mathrm{K}\mathrm{K}^\mathrm{T} = (KR)(KR)^T = M'M'^T = (\mathrm{M}_i\mathrm{A})(\mathrm{M}_i\mathrm{A})^T = \mathrm{M}_i\mathrm{A}\mathrm{A}^\mathrm{T}\mathrm{M}_i{}^\mathrm{T}$$

$$\rightarrow$$

$$\mathrm{M}_i{}^{-1}\omega_\mathrm{i}{}^{-1}\mathrm{M}_i{}^{-\mathrm{T}} = \mathrm{A}\mathrm{A}^\mathrm{T} \qquad \text{Q.E.D}$$

Then, the new i-th camera is $\mathrm{P}_i{}' = \mathrm{P}_i\mathrm{H}^{-1} = [\mathrm{M}_i\mathrm{A}|\mathrm{m}_i]$

and new reconstruction is $\mathrm{X}' = \mathrm{H}\mathrm{X} = \begin{bmatrix} \mathrm{A}^{-1} & 0 \\ 0 & 1 \end{bmatrix}\mathrm{X}$

Affine to metric: the new (other) camera $\mathrm{P}_2{}'$

From $\mathrm{P}_j = [\mathrm{M}_j|\mathrm{m}_j]$, new camera is $\mathrm{P}_j{}' = \mathrm{P}_j\mathrm{H}^{-1} = [\mathrm{M}_j\mathrm{A}|\mathrm{m}_j]$

(since $\mathrm{H} = \begin{bmatrix} \mathrm{A}^{-1} & 0 \\ 0 & 1 \end{bmatrix}$) with $\mathrm{A}\mathrm{A}^{\mathrm{T}} = \left(\mathrm{M}_i{}^{\mathrm{T}}\omega_{\mathrm{i}}\mathrm{M}_i\right)^{-1}$

The ω matrix of the other camera is

$$\omega_j^{-1} = (\mathrm{M}_j\mathrm{A})(\mathrm{M}_j\mathrm{A})^T = \mathrm{M}_j\mathrm{A}\mathrm{A}^{\mathrm{T}}\mathrm{M}_j{}^{\mathrm{T}} = \mathrm{M}_j\left(\mathrm{M}_i{}^{\mathrm{T}}\omega_{\mathrm{i}}\mathrm{M}_i\right)^{-1}\mathrm{M}_j{}^{\mathrm{T}} =$$
$$= \mathrm{M}_j\ \mathrm{M}_i{}^{-1}\ \omega_i^{-1}\ \mathrm{M}_i{}^{-\mathrm{T}}\ \mathrm{M}_j{}^{\mathrm{T}} = \mathrm{H}_\infty\ \omega_i^{-1}\ \mathrm{H}_\infty{}^{\mathrm{T}}$$

Transfer of the ω matrix:

$$\omega_{\mathrm{j}} = \mathrm{H}_\infty{}^{-\mathrm{T}}\omega_{\mathrm{i}}\ \mathrm{H}_\infty{}^{-1}$$

through the homography H_∞ between images *i* and *j* induced by the plane at the ∞

Affine to metric

If, after affine reconstruction, K is known for camera $\mathrm{P}_1 = [\mathrm{I}\,|0\,]$, then this camera is transformed to $\mathrm{P'}_1 = [\mathrm{K}\,|0\,]$ by applying the reconstructing mapping

$$\mathrm{H} = \begin{bmatrix} \mathrm{K}^{-1} & 0 \\ 0 & 1 \end{bmatrix}$$

the other cameras P_2, P_3… are mapped to $\mathrm{P'}_2$, $\mathrm{P'}_3$ …

$$\mathrm{P'}_2 = \mathrm{P}_2\mathrm{H}^{-1} = [\mathrm{M}_2|\mathrm{m}_2]\begin{bmatrix} \mathrm{K} & 0 \\ 0 & 1 \end{bmatrix} = [\mathrm{M}_2\mathrm{K}|\mathrm{m}_2]$$

while the updated (metric) reconstruction is

$$\mathrm{X''} = \mathrm{HX'} = \begin{bmatrix} \mathrm{K}^{-1} & 0 \\ 0 & 1 \end{bmatrix}\mathrm{X'}$$

Constraints on ω from ortogonality

vanishing points corresponding to orthogonal directions

$$v_1^T \omega v_2 = 0$$

vanishing line and vanishing point corresponding to plane and normal direction

$$l = \omega v$$

Constraints on ω from known internal parameters

$$\omega = \mathrm{K}^{-T}\mathrm{K}^{-1}$$

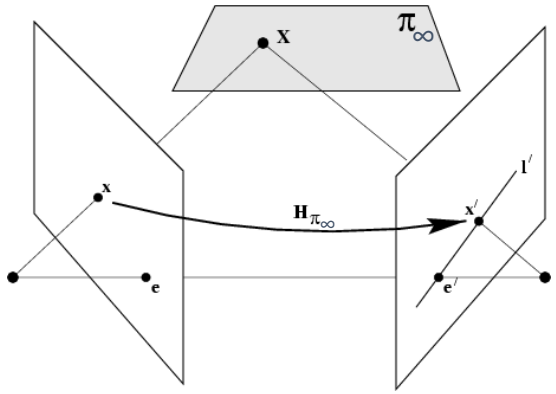
rectangular pixels (zero skew factor)

$$s = 0 \qquad \omega_{12} = \omega_{21} = 0$$

+ square pixels (or known aspect ratio)

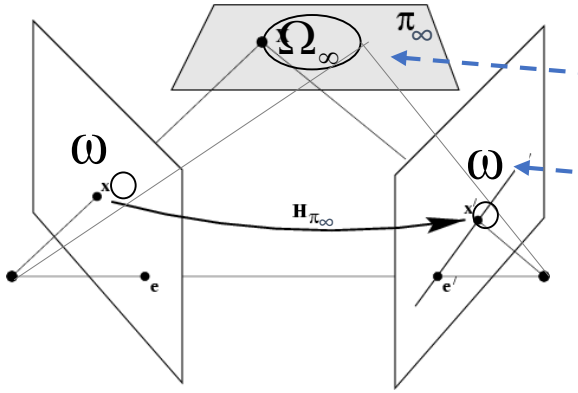
$$f_x = f_y \qquad \omega_{11} = \omega_{22}$$

Remember: **Homography induced by plane at the infinity**



$$H_\infty = M_2 M_1^{-1}$$

transfer of vanishing point: $$v_2 = H_\infty v_1 = M_2 M_1^{-1} v_1$$



absolute conic

IAC image of the absolute conic

transfer of the ω matrix $$\omega_2 = H_\infty^{-T} \omega_1 H_\infty^{-1}$$

Example: same camera for all images
→ constraints on ω from affine reconstruction (H_∞)

same intrinsics $K \Rightarrow$ same ω matrix, e.g. moving camera

$$\omega = \omega' = {H_\infty}^{-T} \omega {H_\infty}^{-1}$$

given enough images there is in general only one matrix ω that transfers to itself in all images,

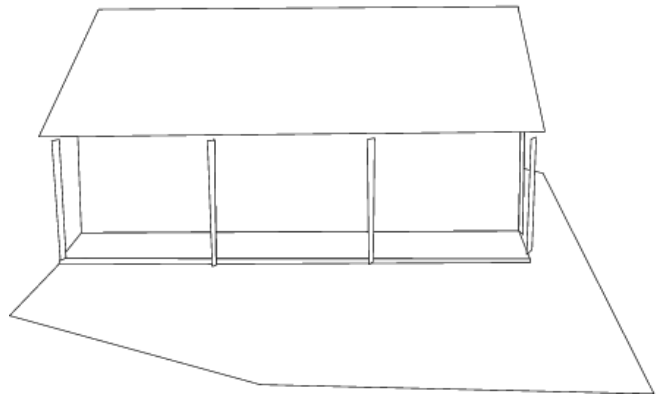
This approach is called *self-calibration,* see later

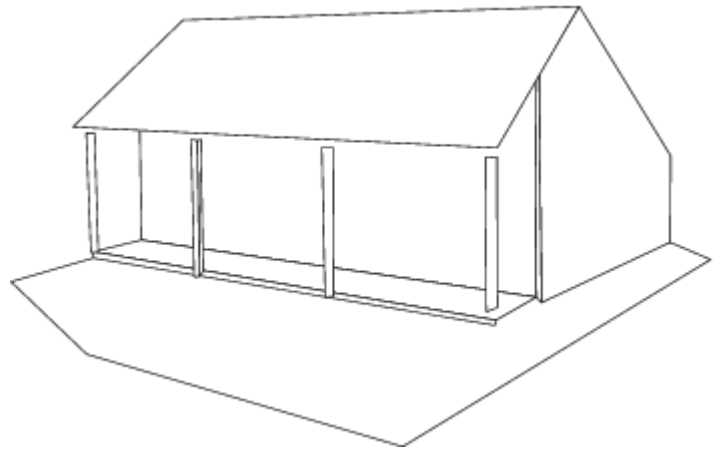
transfer of ω matrix : $\omega' = \omega = {H_\infty}^{-T} \omega \, {H_\infty}^{-1}$

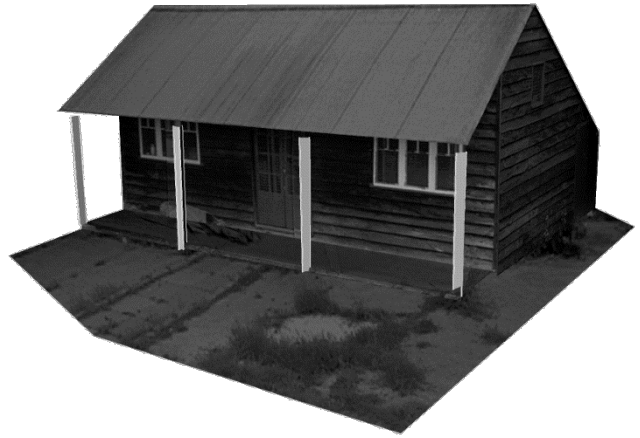
Direct metric reconstruction using ω of all cameras

$$\omega = K^{-T}K^{-1} \Rightarrow K$$

calibrated reconstruction: Essential matrix
→ first relative pose and then stereo triangulation





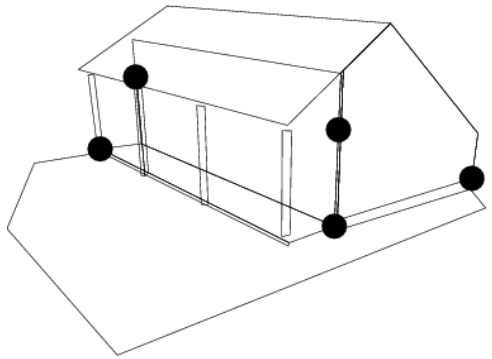




Direct reconstruction using ground truth

use control points X_{Ei} with known coordinates
to go from projective to metric

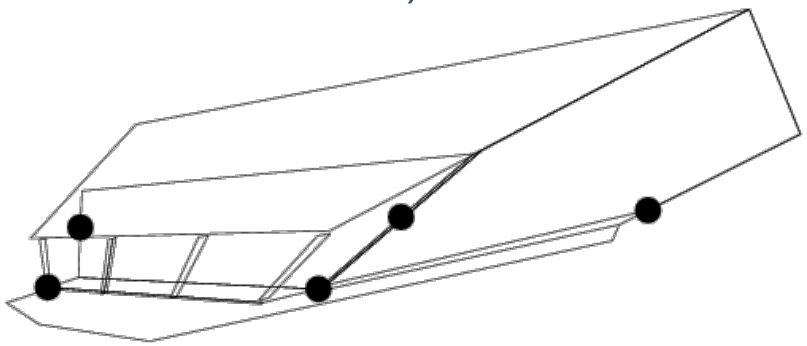




$$X_{Ei} = HX_i$$

$$x_i = PH^{-1}X_{Ei}$$

(2 lin. eq. in H^{-1} per view,
3 for two views)



3D Shape Reconstruction

Data

Enough pairs of corresponding image points

$$\mathrm{x}_{1i} \leftrightarrow \mathrm{x}_{2i}$$

taken by two uncalibrated images

Purpose

Compute

$$(\mathrm{P}_1^{\mathrm{M}}, \mathrm{P}_2^{\mathrm{M}}, \{\mathrm{X}_{\mathrm{i}}^{\mathrm{M}}\})$$

(i.e. within similarity of original scene and cameras)

Algorithm

(i) Compute projective reconstruction $(\mathrm{P}_1^{\mathrm{o}}, \mathrm{P}_2^{\mathrm{o}}, \{\mathrm{X}_{\mathrm{i}}^{\mathrm{o}}\})$

(a) Compute F from $\mathrm{x}_{1i} \leftrightarrow \mathrm{x}_{2i}$

(b) Compute $\mathrm{P}_1^{\mathrm{o}}, \mathrm{P}_2^{\mathrm{o}}$ from F: $\mathrm{P}_1^{\mathrm{o}} = [\mathrm{I}|0]$ $\mathrm{P}_2^{\mathrm{o}} = [[\mathrm{e}_2]_{\times} F \mid \mathrm{e}_2]$, ($\mathrm{e}_2$=LNS F)

(c) Triangulate X_{i} from $\mathrm{x}_{1i} \leftrightarrow \mathrm{x}_{2i}$

(ii) Rectify reconstruction from projective to metric

Direct method: compute H from control points $\mathrm{X}_{\mathrm{E}i} = \mathrm{H}\mathrm{X}_i$

then $\mathrm{P}_1^{\mathrm{M}} = \mathrm{P}_1^{\mathrm{o}}\mathrm{H}^{-1}, \mathrm{P}_1^{\mathrm{M}} = \mathrm{P}_1^{\mathrm{o}}\mathrm{H}^{-1}, \mathrm{X}_{\mathrm{E}i} = \mathrm{H}\mathrm{X}_i$

Stratified method:

(a) **Affine reconstruction**: find (fit) π_∞ and apply $\mathrm{H} = \begin{bmatrix} \mathrm{I}|0 \\ \pi_\infty \end{bmatrix}$ **OR**

find (fit) H_∞ and set new cameras $\mathrm{P}_1^{\mathrm{A}} = [\mathrm{I}|0]$ and $\mathrm{P}_2^{\mathrm{A}} = [\mathrm{H}_\infty|\mathrm{e}_2]$

(a) **Metric reconstruction**: find K → (ω matrix) for one of the cameras

apply $\mathrm{H} = \begin{bmatrix} \mathrm{A}^{-1} & 0 \\ 0 & 1 \end{bmatrix}$ with $\mathrm{A}\mathrm{A}^{\mathrm{T}} = \left(\mathrm{M}^{\mathrm{T}}\omega\mathrm{M}\right)^{-1}$ Cholesky factorisation

Image information provided	View relations and projective objects	3-space objects	reconstruction ambiguity
point correspondences	F		projective
point correspondences including vanishing points	F, H_∞	π_∞	affine
Points correspondences and internal camera calibration	F, H_∞ ω, ω'	π_∞ Q_∞	metric

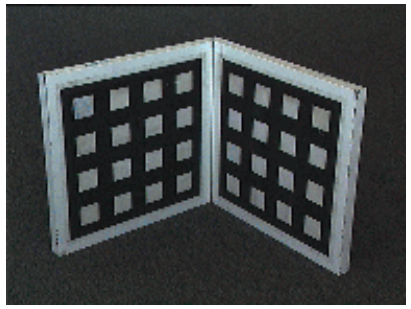
SELF-CALIBRATION

Outline

- Introduction
- Self-calibration
- Absolute Conic

Motivation

- Avoid explicit calibration procedure
  - Complex procedure
  - Need for calibration object
  - Need to maintain calibration



Motivation

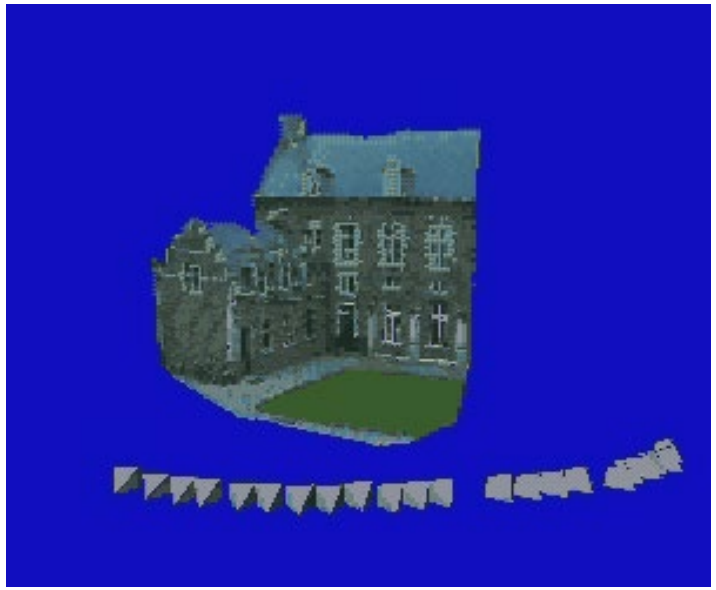
- Allow flexible acquisition
  - No prior calibration necessary
  - Possibility to vary intrinsics
  - Use archive footage

Projective ambiguity

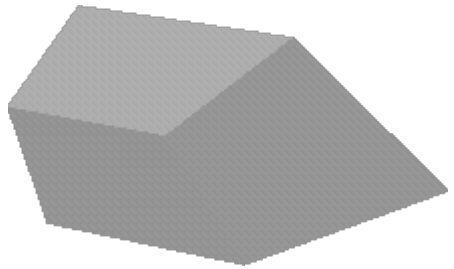
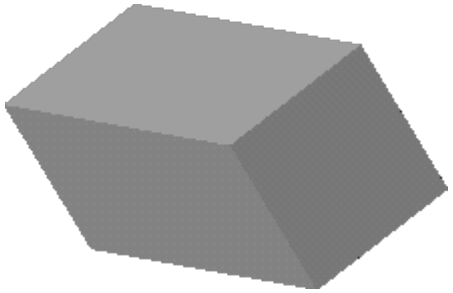
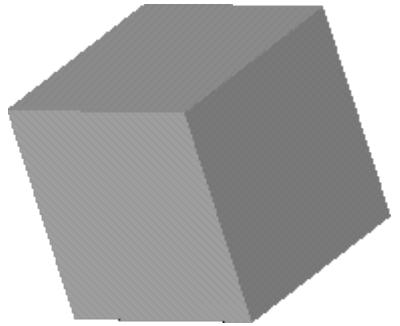
Reconstruction from uncalibrated images

$\Rightarrow$ projective ambiguity on reconstruction

$$\mathrm{x} = \mathbf{P}\mathrm{X} = (\mathbf{P}\mathbf{T}^{-1})(\mathbf{T}\mathrm{X}) = \mathbf{P}'\mathrm{X}'$$



Stratification of geometry

Projective	Affine	Metric
		
15 DOF	12 DOF *plane at infinity* parallelism	7 DOF *absolute conic* angles, rel.dist.

More general

More structure

Constraints ?

- Scene constraints
  - Parallellism, vanishing points, horizon, ...
  - Distances, positions, angles, ...

  Unknown scene → no constraints
- Camera extrinsics constraints
  - Pose, orientation, motion ...

  Unknown camera motion → no constraints
- Camera intrinsics constraints
  - Focal length, principal point, aspect ratio & skew

  Perspective camera model too general
  → some constraints

Euclidean projection matrix

Factorization of Euclidean projection matrix

$$\mathbf{P} = \mathbf{K}\begin{bmatrix}\mathbf{R}^{\mathsf{T}} & -\mathbf{R}^{\mathsf{T}}\mathbf{t}\end{bmatrix}$$

Intrinsics: $\mathbf{K} = \begin{bmatrix} f_x & s & u_x \\ & f_y & u_y \\ & & 1 \end{bmatrix}$ (camera geometry)

Extrinsics: $(\mathbf{R}, \mathrm{t})$ (camera motion)

Note: every projection matrix can be factorized,
but only meaningful for euclidean projection matrices

Self-calibration

Upgrade from *projective* structure to *metric* structure using *constraints on intrinsic* camera parameters

- Constant intrinsics

(Faugeras et al. ECCV´92, Hartley´93, Triggs´97, Pollefeys et al. PAMI´98, ...)

- Some known intrinsics, others varying

(Heyden&Astrom CVPR´97, Pollefeys et al. ICCV´98,...)

- Constraints on intrinics and restricted motion
  (e.g. pure translation, pure rotation, planar motion)

(Moons et al.´94, Hartley ´94, Armstrong ECCV´96, ...)

A counting argument

- To go from projective (15DOF) to metric (7DOF) at least 8 constraints are needed
- Minimal sequence length should satisfy

$$n \times (\#\,known) + (n-1) \times (\#\,fixed) \geq 8$$

where n is the number of images

- Independent of algorithm
- Assumes general motion (i.e. not critical)

Dual Image of Absolute Conic DIAC

From tentative cameras to true cameras

tentative cameras $P_1 = [I \quad 0] \qquad P_i = [A_i \quad a_i]$

true cameras $P_1' = P_1 H^{-1} \quad P_i' = P_i H^{-1}$

with $H^{-1} = \begin{bmatrix} A & t \\ v^T & 1 \end{bmatrix}$

world reference = first-camera reference

$$P_1' = [K_1 \quad 0] = [I \quad 0] \begin{bmatrix} A & t \\ v^T & 1 \end{bmatrix}$$

thus $A = K_1$ and $t = 0$

.. therefore

new camera $P_i' = [K_i R_i \quad K_i t_i] = [A_i \quad a_i] \begin{bmatrix} K_1 & 0 \\ v^T & 1 \end{bmatrix}$

now define $p \quad s.t. \quad v^T = -p^T K_1$

then the new camera $P_i' = [(A_i - a_i p^T) K_1 \quad a_i]$

is a true camera if $(A_i - a_i p^T) K_1 = K_i R_i$

But, since $\omega_i^* = K_i K_i^T = (K_i R_i)(K_i R_i)^T$

Then $\omega_i^* = (A_i - a_i p^T) K_1 K_1^T (A_i - a_i p^T)^T$

Hence $\omega_i^* = (A_i - a_i p^T) \omega_1^* (A_i - a_i p^T)^T$

Selfcalibration equation

$$\omega_i^* = (A_i - a_i p^T)\omega_1^*(A_i - a_i p^T)^T$$

8 unknowns: p and ω_1^* (3 + 5)

provided that only constrained elements of ω_i^* are used (i.e., known or constant)

→ in this way: new equations do not introduce new unknowns

$$\omega_i^* = (A_i - a_i p^T)\, \omega_1^*\, (A_i - a_i p^T)^T$$

$$A_i - a_i p^T = \mathrm{H}_{\infty i}$$

since true cameras are $[K_1|0]$, $[(A_i - a_i p^T)K_1|a_i]$
and the homography induced by the plane π_∞ is

$$\mathrm{H}_{\infty i} = \mathrm{M}_2 {\mathrm{M}_1}^{-1} = A_i - a_i p^T$$

there are 8 unknowns: vector p and matrix K_1

Constraints on intrinsic parameters

$$\mathbf{K} = \begin{bmatrix} f_x & s & u_x \\ & f_y & u_y \\ & & 1 \end{bmatrix}$$

Constant

same cameras: $\mathbf{K}_1 = \mathbf{K}_2 = \cdots$

aspect ratio $f_x / f_y = a$ constant for all cameras

Known

e.g. rectangular pixels: $s = 0$

square pixels: $f_x = f_y, s = 0$

principal point known: $(u_x, u_y) = \left(\frac{w}{2}, \frac{h}{2}\right)$

A counting argument

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- Minimal sequence length should satisfy

$$n \times (\#\,known) + (n-1) \times (\#\,fixed) \geq 8$$

- Independent of algorithm
- Assumes general motion (i.e. not critical)

Constraints on DIAC ω^*

$$\omega^* = \begin{bmatrix} f_x^2 + s^2 + u_o^2 & sf_y + u_o v_o & u_o \\ sf_y + u_o v_o & f_y^2 + v_o^2 & v_o \\ u_o & v_o & 1 \end{bmatrix}$$

condition	constraint	type	#constraints
Zero skew	$\omega^*_{12}\omega^*_{33} = \omega^*_{13}\omega^*_{23}$	quadratic	m
Principal point	$\omega^*_{13} = \omega^*_{23} = 0$	linear	$2m$
Zero skew (& p.p.)	$\omega^*_{12} = 0$	linear	m
Fixed aspect ratio (& p.p.& Skew)	$\omega^*_{11}\omega'^*_{22} = \omega^*_{22}\omega'^*_{11}$	quadratic	m-1
Known aspect ratio (& p.p.& Skew)	$\omega^*_{11} = \omega^*_{22}$	linear	m
Focal length (& p.p. & Skew)	$\omega^*_{33} = \omega^*_{11}$	linear	m

Note that in the absence of skew the IAC ω can be more practical than the DIAC ω^*!

$$\omega_\infty^* = \begin{bmatrix} f_x^2 + s^2 + u_o^2 & sf_y + u_o v_o & u_o \\ sf_y + u_o v_o & f_y^2 + v_o^2 & v_o \\ u_o & v_o & 1 \end{bmatrix}$$

$$\omega = \frac{1}{f_x^2 f_y^2} \begin{bmatrix} f_y^2 & 0 & -f_y^2 u_o \\ 0 & f_x^2 & -f_x^2 v_o \\ -f_y^2 v_o & -f_x^2 u_o & f_x^2 f_y^2 + f_y^2 u_o^2 + f_x^2 v_o^2 \end{bmatrix}$$

Linear algorithm

(Pollefeys et al.,ICCV´98/IJCV´99)

Assume everything known, except focal length

$$\omega^* \approx \begin{bmatrix} \hat{f}^2 & 0 & 0 \\ 0 & \hat{f}^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \propto \mathbf{P}\Omega^*\mathbf{P}^T$$

$$\left(\mathbf{P}\mathbf{Q}^*\mathbf{P}^T\right)_{11} - \left(\mathbf{P}\mathbf{Q}^*\mathbf{P}^T\right)_{22} = 0$$
$$\left(\mathbf{P}\mathbf{Q}^*\mathbf{P}^T\right)_{12} = 0$$
$$\left(\mathbf{P}\mathbf{Q}^*\mathbf{P}^T\right)_{13} = 0$$
$$\left(\mathbf{P}\mathbf{Q}^*\mathbf{P}^T\right)_{23} = 0$$

Yields 4 constraint per image

Note that rank-3 constraint is not enforced

Linear algorithm revisited

(Pollefeys et al., ECCV'02)

Weighted linear equations

$$\mathbf{K}\mathbf{K}^{\mathsf{T}} \approx \begin{bmatrix} \hat{f}^2 & 0 & 0 \\ 0 & \hat{f}^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\hat{f} \approx 1$$

$$\frac{1}{0.2}\left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{11} - \left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{22} = 0$$

$$\frac{1}{0.01}\left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{12} = 0$$

$$\frac{1}{0.1}\left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{13} = 0$$

$$\frac{1}{0.1}\left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{23} = 0$$

$$\frac{1}{9}\left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{11} - \left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{33} = 0$$

$$\frac{1}{9}\left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{22} - \left(\mathbf{P}\Omega^*\mathbf{P}^{\mathsf{T}}\right)_{33} = 0$$

assumptions

$$\log(\hat{f}) \approx \log(1) \pm \log(3)$$

$$\log\left(\frac{\hat{f}_x}{\hat{f}_y}\right) \approx \log(1) \pm \log(1.1)$$

$$c_x \approx 0 \pm 0.1 \qquad s = 0$$

$$c_y \approx 0 \pm 0.1$$

Refinement

- Metric bundle adjustment

$$\underset{\mathbf{P}_k, \mathrm{X}_i}{\arg\min} \sum_{k=1}^{m} \sum_{i=1}^{n} D\left(\mathrm{x}_{ki}, \mathbf{P}_k\left(\mathrm{X}_i\right)\right)^2$$

Enforce constraints or priors
on intrinsics during minimization
(this is „self-calibration“ for photogrammetrist)